

An off-shell conformally invariant Galilean Weyl tensor

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We propose a manifestly conformally invariant off-shell definition of the Weyl tensor in Galilean geometry. We also propose definitions for the electric and magnetic parts thereof. For the latter to vanish, observers with specific kinematical properties need to exist. While this is guaranteed by the Newton–Cartan equation, it might not be true for other Galilean invariant theories. Therefore, we propose that imposing the existence of observers for which the off-shell magnetic part is zero should be a necessary condition for a Galilean invariant theory to be called ‘Newtonian’, in the spirit of the ‘Newtonian’ condition, introduced by Trautman, in standard Newton–Cartan gravity.

As a side result, we also show that there exists a unique Galilean boost-invariant connection that can be built from a Galilean structure and a choice of Coriolis field, even when the clock form is not closed. No extra structure is needed, contrasting with the standard approach used to construct boost-invariant connections which introduces a mass gauge field.

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1. Introduction

In relativistic (Lorentzian) spacetime physics, a central object of study is the Weyl tensor—i.e., the trace-free part of the Riemann tensor. There are several (related) reasons for this: (i) the Weyl tensor is conformally invariant and thus carries important information about the conformal structure of spacetime; (ii) the Weyl tensor encodes the gravitational degrees of freedom which are retained even in vacuum solutions of general relativity—i.e., even in the absence of matter (when $R_{\mu\nu} = 0$); (iii) the Weyl tensor is a crucial object of study in the physics of gravitational waves; (iv) the Weyl tensor can be used to facilitate a classification of relativistic spacetimes (the famous Petrov classification); and so forth. All of this is standard lore, and can be found in any good textbook treatment of general relativity.

What has only recently begun to arouse interest, however, is how key structures in relativistic physics (including, but not limited to, the Weyl tensor) carry over to and behave in theories of spacetime and gravitation other than general relativity. For example, the question how the notion of conformal structure carries over to the context of Galilean gravity (i.e., ‘non-relativistic’ gravity—roughly, the $c \rightarrow \infty$ limit of general relativity [1–12]) and Carrollian gravity (i.e., ‘ultra-relativistic’ gravity—roughly, the $c \rightarrow 0$ limit of general relativity [13–16]) has been picked up in refs. [17, 18], and sensible definitions of these non-Lorentzian conformal structures have been proposed therein.

One might think that exploring the status of such structures in spacetime geometries other than the Lorentzian setting is a mere academic exercise, since (of course) our actual universe is relativistic. However, to say this would be too fast: on the one hand, Galilean gravity has recently found wide application in *inter alia* condensed matter phenomena (especially the fractional quantum Hall effect, see refs. [19–21]), Hořava-Lifshitz gravity [22], non-relativistic holography [23], and non-relativistic string theory [24]; on the other hand, Carrollian gravity has recently become *à la mode* in fields such as flat-space holography [25] and the physics of black holes (see, e.g., ref. [26])—the latter often via Carrollian string theory, on which see, e.g., ref. [27]. Given the recent wide application of these non-relativistic geometries, there is a good case to be made that the project of seeking to understand the structures and invariants related to them is a worthy one.

As such, in this article we pick up on the notion of conformal structure in the Galilean context (the Carrollian context being set aside for another day), and in particular explore the Weyl tensor—*qua* conformal invariant—in that setting. Although there is already some writing on this topic—see in particular refs. [28–30]—that work sometimes involves mathematical errors, and in any case is not maximally general insofar as the results obtain only on-shell—i.e., when the equations of motion are imposed.

With this in mind, our plan for the current paper is as follows. In Section 2, we recall the definitions of the Weyl tensor for Lorentzian structures, and its electric and magnetic parts. In Section 3, we discuss Galilean structures, their conformal transformations, and what affine connections should be considered given such a structure. Section 4 presents the Galilean limit of Lorentzian geometry. Then, applying this limit to the Lorentzian

Weyl tensor, along with the material introduced in [Section 3](#), we provide in [Section 5](#) a conformally invariant off-shell definition for the Galilean Weyl tensor and its electric and magnetic parts. We discuss the significance of these definitions in [Section 6](#), and we conclude in [Section 7](#).

1.1. Notations

We will employ the following notation:

- The coordinate components of a connection are given by $\nabla_\mu v^\nu =: \partial_\mu v^\nu + \Gamma_{\mu\sigma}^\nu v^\sigma$.
- Unit timelike vector fields with respect to a Lorentzian structure will be denoted by uppercase letters (e.g., U^μ); unit timelike vector fields with respect to a Galilean structure will be denoted by lowercase letters (e.g., u^μ). Some tensors will depend on the choice of timelike vector field, e.g., the Coriolis field $\kappa_{\mu\nu}$. These tensors should feature an upper-left index ‘ u ’ indicating this dependence. However, unless an ambiguity can arise, we will drop this index after the introduction of the tensor.
- The speed of light is denoted by c and is not assumed to be unity, in order to properly perform the Galilean limit.¹ We write $\lambda := 1/c^2$.
- In the context of the Galilean limit, we will write ‘LO’ for ‘leading order’.
- The symmetric traceless part of a two-tensor field will be denoted by enclosing the corresponding two indices in angular brackets. The tracelessness is defined with respect to that non-degenerate metric which is clear from context, which in our case will be the spatial metric from either the Lorentzian or the Galilean structures. For example, in three dimensions, denoting the metric by h_{ij} , we write $A_{\langle ij \rangle} := A_{(ij)} - \frac{1}{3} h^{kl} A_{kl} h_{ij}$.

2. The Weyl tensor of Lorentzian structures

In this section we recall, for Lorentzian structures, the definitions of the Weyl tensor, its electric and magnetic parts, the 1+3 formalism, and conformal transformations.

The Weyl tensor: Given a Lorentzian metric $g_{\mu\nu}$, its Weyl tensor is defined as

$$C^\alpha{}_{\beta\mu\nu} := R^\alpha{}_{\beta\mu\nu} - \delta_{[\mu}^\alpha R_{\nu]\beta} + g_{\beta[\mu} R_{\nu]\sigma} g^{\sigma\alpha} + \frac{1}{3} \delta_{[\mu}^\alpha g_{\nu]\beta} R_{\sigma\lambda} g^{\sigma\lambda}. \quad (2.1)$$

Given a unit timelike vector field U^μ , i.e., $U_\sigma U^\sigma = -1/\lambda$, and its orthogonal projector ${}^U b_{\mu\nu} := g_{\mu\nu} + \lambda U_\mu U_\nu$, the electric part ${}^U E_{\mu\nu}$ and magnetic part ${}^U H_{\mu\nu}$ of the Weyl tensor

¹We use the terminology ‘Galilean’ rather than ‘Newtonian’ for denoting the $c \rightarrow \infty$ limit of Lorentzian geometry to highlight the fact that we are in a geometric (off-shell) situation.

with respect to U^μ are defined as

$$\begin{aligned} {}^U E_{\mu\nu} &:= C_{\mu\alpha\nu\beta} U^\alpha U^\beta \\ &= \frac{1}{\lambda} \frac{1}{2} b^\beta{}_\mu b^\alpha{}_\nu \left(R_{\alpha\beta} - \frac{1}{3} b_{\alpha\beta} R_{\sigma\omega} g^{\sigma\omega} \right) - \frac{1}{2} b_{\mu\nu} R_{\alpha\beta} U^\alpha U^\beta + R_{\mu\alpha\nu\beta} U^\alpha U^\beta, \end{aligned} \quad (2.2)$$

$$\begin{aligned} {}^U H_{\mu\nu} &:= \frac{1}{2} {}^U \epsilon_{\mu\alpha\beta} C^{\alpha\beta}{}_{\nu\sigma} U^\sigma \\ &= \frac{1}{2} {}^U \epsilon_{\alpha\beta(\mu} R^{\alpha\beta}{}_{\nu)\sigma} U^\sigma, \end{aligned} \quad (2.3)$$

where ${}^U \epsilon_{\beta\mu\nu} := \epsilon_{\alpha\beta\mu\nu} U^\alpha$ and $\epsilon_{\alpha\beta\mu\nu}$ is the spacetime volume form associated to $g_{\mu\nu}$. The Weyl tensor can be expressed in terms of the electric and magnetic parts as

$$C_{\mu\nu}{}^{\alpha\beta} = \lambda 4 \left(b^{[\alpha}{}_{[\mu} E^{\beta]}{}_{\nu]} + \lambda U^{[\alpha} U_{[\mu} E^{\beta]}{}_{\nu]} \right) + \lambda 2 \epsilon_{\mu\nu\sigma} U^{[\alpha} H^{\beta]\sigma} + \lambda 2 U_{[\mu} H_{\nu]\sigma} \epsilon^{\alpha\beta\sigma}. \quad (2.4)$$

The 1+3 formalism: Using the 1+3 formalism, it is possible to express the electric and magnetic parts in terms of quantities purely orthogonal² to U^μ , called *kinematical quantities*. We describe this approach in the following. For a complete introduction on the 1+3 formalism, we direct the reader to ref. [31].

The kinematical (1+3) quantities of U^μ are defined as follows:

$$\text{the acceleration:} \quad {}^U a^\mu := U^\alpha \nabla_\alpha U^\mu, \quad (2.5)$$

$$\text{the expansion tensor:} \quad {}^U \Theta_{\mu\nu} := b^\alpha{}_{(\mu} b^\beta{}_{\nu)} \nabla_\alpha U_\beta = \frac{1}{2} \mathcal{L}_U b_{\mu\nu}, \quad (2.6)$$

$$\text{the vorticity tensor:} \quad {}^U \Omega_{\mu\nu} := b^\alpha{}_{[\mu} b^\beta{}_{\nu]} \nabla_\alpha U_\beta, \quad (2.7)$$

$$\text{the expansion rate:} \quad {}^U \theta := \Theta^\alpha{}_\alpha, \quad (2.8)$$

$$\text{the shear tensor:} \quad {}^U \sigma_{\mu\nu} := \Theta_{\langle\mu\nu\rangle}, \quad (2.9)$$

where $\langle\mu\nu\rangle$ denotes the traceless part with respect to the spatial projector $b_{\mu\nu}$, i.e., $\Theta_{\langle\mu\nu\rangle} := \Theta_{\mu\nu} - \frac{1}{3} \theta b_{\mu\nu}$.

We define the ‘spatial connection’ ${}^U D_\mu$ acting on tensor fields $A^{\mu\dots}{}_{\nu\dots}$ orthogonal to U^μ as the operator

$${}^U D_\alpha A^{\mu\dots}{}_{\nu\dots} := b^\omega{}_\alpha b^\mu{}_\sigma \dots b^\lambda{}_\nu \dots \nabla_\omega A^{\sigma\dots}{}_{\lambda\dots}. \quad (2.10)$$

Following Section 5.1.1 in ref. [31], we define the ‘spatial Riemann tensor’ ${}^U \mathcal{R}^\alpha{}_{\beta\mu\nu}$ as

$$\omega_\alpha w^\beta v_1^\mu v_2^\nu {}^U \mathcal{R}^\alpha{}_{\beta\mu\nu} := \omega_\alpha \left[(D_{v_1} D_{v_2} w - D_{v_2} D_{v_1} w)^\alpha - b^\alpha{}_\sigma (\nabla_{[v_1, v_2]} w)^\sigma \right], \quad (2.11)$$

where ω^μ , w^μ , v_1^μ , and v_2^μ are all orthogonal to u^μ .

²Since we may freely identify the tangent and cotangent spaces via the non-degenerate Lorentzian metric $g_{\mu\nu}$, the notion of being orthogonal to U^μ is well-defined for both vectors and covectors, and hence for (indices of) both contra- and covariant tensors.

Remark 2.1. If the vorticity vanishes, i.e., U_μ is foliation forming, then D_μ is the Levi-Civita connection of the spatial Riemannian metric induced by $g_{\mu\nu}$ on each of the spatial leaves integrating the distribution $\ker U_\mu$.

If the vorticity does *not* vanish, $\Omega_{\mu\nu} \neq 0$, the operator D_μ may *not* be understood as a vector bundle connection (covariant derivative operator): it does not satisfy the Leibniz rule for differentiation in direction of U^μ (since this yields 0 by definition), so it cannot be a connection on the vector bundle $\ker U_\mu$ over the spacetime manifold; further, since no spatial leaves exist, it cannot be understood as an affine connection on these (i.e., as a covariant derivative on the leaves' tangent bundles). In particular, it cannot be understood as the Levi-Civita connection of a 'Riemannian spatial metric'. In particular, its 'torsion', defined by

$$T[D]^\alpha{}_{\mu\nu} \omega_\alpha v_1^\mu v_2^\nu := \omega_\alpha [(D_{v_1} w - D_{v_2} w)^\alpha - [v, w]^\alpha] \quad (2.12)$$

for $\omega^\mu, v_1^\mu, v_2^\mu$ orthogonal to u^μ (see Section 4.2.3 in ref. [31]), is not purely spatial and given by

$$T[D]^\alpha{}_{\mu\nu} = -\lambda \, 2 \, \Omega_{\mu\nu} U^\alpha. \quad (2.13)$$

In particular, the presence of this torsion leads to an antisymmetric part of the 'spatial Ricci tensor' (defined by contraction of $\mathcal{R}^\alpha{}_{\beta\mu\nu}$):

$$\mathcal{R}_{[\mu\nu]} = \lambda [\theta \Omega_{\mu\nu} + 2\Theta^\alpha{}_{[\mu} \Omega_{\nu]\alpha}]. \quad (2.14)$$

Therefore, this tensor does not enjoy the properties of the Ricci tensor defined from a Riemannian metric.

However, as will be shown in Section 4.2, at leading order in the Galilean limit, D_μ corresponds to the Levi-Civita connection of the Galilean spatial (Riemannian) metric, and the Ricci tensor is symmetric.

With respect to the kinematical variables, the electric part of the Weyl tensor takes the form (see Section 4.7.2 in ref. [32])

$$E_{\mu\nu} = \frac{1}{\lambda} \frac{1}{2} R_{\alpha\beta} b^\alpha{}_{\langle\mu} b^\beta{}_{\nu\rangle} - \mathcal{L}_U \sigma_{\langle\mu\nu\rangle} + \sigma_{\alpha\langle\mu} \sigma_{\nu\rangle}{}^\alpha + \Omega_{\alpha\langle\mu} \Omega_{\nu\rangle}{}^\alpha + 2\sigma_{\alpha\langle\mu} \Omega_{\nu\rangle}{}^\alpha + D_{\langle\mu} a_{\nu\rangle} + \lambda a_{\langle\mu} a_{\nu\rangle}. \quad (2.15)$$

We want to express the first term in that formula in terms of the spatial curvature. By Corollary S3 in ref. [31], we have

$$\frac{1}{\lambda} R_{\alpha\beta} b^\alpha{}_{\langle\mu} b^\beta{}_{\nu\rangle} = \mathcal{R}_{\langle\mu\nu\rangle} + \mathcal{L}_U \sigma_{\langle\mu\nu\rangle} + \frac{1}{3} \theta \sigma_{\mu\nu} - 2\sigma^\alpha{}_{\langle\mu} \sigma_{\nu\rangle\alpha} - 2\sigma_{\alpha\langle\mu} \Omega_{\nu\rangle}{}^\alpha - D_{\langle\mu} a_{\nu\rangle} - \lambda a_{\langle\mu} a_{\nu\rangle}. \quad (2.16)$$

Then, the electric part (2.15) can be expressed in terms of the spatial curvature as

$$E_{\mu\nu} = \frac{1}{2} \left[\frac{1}{\lambda} \mathcal{R}_{\langle\mu\nu\rangle} - \mathcal{L}_U \sigma_{\langle\mu\nu\rangle} + \frac{1}{3} \theta \sigma_{\mu\nu} + 2 \left(\Theta_{\alpha\langle\mu} + \Omega_{\alpha\langle\mu} \right) \Omega_{\nu\rangle}{}^\alpha + D_{\langle\mu} a_{\nu\rangle} + \lambda a_{\langle\mu} a_{\nu\rangle} \right]. \quad (2.17)$$

The magnetic part is (see Section 4.7.2 in ref. [32])

$$H_{\mu\nu} = \epsilon_{\alpha\beta\langle\mu} [D^\alpha (\sigma^\beta{}_{\nu\rangle} + \Omega^\beta{}_{\nu\rangle}) - \lambda a_{\mu\rangle} \Omega^{\alpha\beta}]. \quad (2.18)$$

Conformal transformations: A conformal transformation of a Lorentzian metric $g_{\mu\nu}$ is defined as

$$g_{\mu\nu} \xrightarrow{\varphi} e^{2\varphi} g_{\mu\nu} , \quad (2.19)$$

where φ is a scalar field. Under this transformation, the 1+3 quantities transform as

$$U^\mu \xrightarrow{\varphi} e^{-\varphi} U^\mu , \quad \theta \xrightarrow{\varphi} e^{-\varphi} (\theta + 3U^\alpha \partial_\alpha \varphi) , \quad \sigma_{\mu\nu} \xrightarrow{\varphi} e^\varphi \sigma_{\mu\nu} , \quad (2.20a)$$

$$\Omega_{\mu\nu} \xrightarrow{\varphi} e^\varphi \Omega_{\mu\nu} , \quad a_\mu \xrightarrow{\varphi} a_\mu + \frac{1}{\lambda} b^\alpha_\mu \partial_\alpha \varphi . \quad (2.20b)$$

The Weyl tensor, its electric part, and its magnetic part are conformally invariant.

3. Galilean structures, connections, and conformal transformations

In this section we recall the definitions of Galilean structures, their conformal transformations, and the decomposition theorem of any affine connection with respect to these structures. Then, we derive the connections which can be constructed from a Galilean structure alone. For the basics of Galilean geometry, see, e.g., refs. [10–12].

3.1. Galilean structures

A *Galilean structure* is a pair $(\tau_\mu, h^{\mu\nu})$ where τ_μ is a non-vanishing 1-form called the *clock form*, and $h^{\mu\nu}$ is a positive-semidefinite symmetric (2,0)-tensor field of rank 3, satisfying $\tau_\mu h^{\mu\nu} = 0$, called the *spatial metric*.

A vector (field) v^μ is said to be *spacelike* if $\tau_\alpha v^\alpha = 0$, and similarly for higher-degree contravariant tensors. A vector (field) u^μ is said to be *unit timelike* with respect to the Galilean structure if $\tau_\alpha u^\alpha = 1$, and (future-directed) *timelike* if $\tau_\alpha u^\alpha > 0$. For the rest of the paper we will always consider unit timelike vector fields, therefore we will drop ‘unit’ for simplicity.

The field $h^{\mu\nu}$ induces a positive-definite bundle metric on the distribution $\ker \tau_\mu$ of spacelike vectors, i.e., a proper ‘spatial metric’; see, e.g., Construction 2.5 in ref. [12].

The *spatial projector* along a timelike vector field u^μ is

$${}^u \gamma^\mu_\nu := \delta^\mu_\nu - u^\mu \tau_\nu . \quad (3.1)$$

The *covariant spatial metric* ${}^u \gamma_{\mu\nu} = {}^u \gamma^\mu_\nu$ with respect to u^μ is (uniquely) defined by

$${}^u \gamma_{\nu\alpha} h^{\alpha\mu} = {}^u \gamma^\mu_\nu , \quad {}^u \gamma_{\mu\alpha} u^\alpha = 0 . \quad (3.2)$$

We use the letter γ rather than b to distinguish these objects from the orthogonal projector defined for Lorentzian structures.

While a Galilean structure alone does not define a dual relation between forms and vectors, a standard convention is to define a (non-invertible) way of raising and lowering

indices via contraction with the spatial metric $h^{\mu\nu}$ and the covariant spatial metric $\gamma_{\mu\nu}$, given a fixed choice of timelike vector field u^μ . In this sense raising indices only depends on the Galilean structure, while lowering indices depends on a choice of timelike vector field. Since for most of this paper that vector field will be fixed, we can raise and lower indices without ambiguity.

We introduce

$$\omega_{\mu\nu} := \frac{1}{2}(\mathrm{d}\tau)_{\mu\nu} = \partial_{[\mu}\tau_{\nu]} . \quad (3.3)$$

For a general Galilean structure, there is no constraint on $\omega_{\mu\nu}$. In particular, in general we have $\tau \wedge \mathrm{d}\tau \neq 0$, implying that the distribution $\ker \tau_\mu$ of spacelike vectors is not integrable. For the present paper we will however only consider clock forms satisfying³

$$\tau \wedge \mathrm{d}\tau = 0 , \quad (3.4a)$$

such that the spacelike distribution is integrable. We call the foliation of the spacetime manifold integrating the spacelike distribution $\ker \tau_\mu$ the *spatial foliation*, or the foliation by *spatial leaves*. The bundle metric induced by $h^{\mu\nu}$ on $\ker \tau_\mu$ is then a (positive-definite) Riemannian metric on each spatial leaf Σ , which we denote by h_{ij} and call the *spatial metric* as well (in the following, no ambiguity will arise regarding whether we speak of $h^{\mu\nu}$ or h_{ij}). On top of complying with most of the literature on Galilean structures, the hypothesis of τ_μ being foliation forming is also imposed by global hyperbolicity if we require the Galilean structure to arise from a Galilean limit of a Lorentzian metric (see [Section 4.1](#) and ref. [\[33\]](#)). The condition $\tau \wedge \mathrm{d}\tau = 0$ is equivalent to

$$h^{\mu\alpha} h^{\nu\beta} \omega_{\alpha\beta} = 0 . \quad (3.4b)$$

Then, choosing any timelike vector field u^μ , $\omega_{\alpha\beta}$ is fully determined by the vector field

$$\mathcal{A}^\mu := 2 u^\alpha \omega_{\alpha\beta} h^{\beta\mu} = h^{\mu\alpha} \mathcal{L}_u \tau_\alpha , \quad (3.4c)$$

with

$$\omega_{\mu\nu} = \tau_{[\mu} \mathcal{A}_{\nu]} := \tau_{[\mu} \gamma_{\nu]\alpha} \mathcal{A}^\alpha . \quad (3.4d)$$

Both $\mathcal{A}^\mu$ and $\tau_{[\mu} \mathcal{A}_{\nu]}$ do not depend on the choice of timelike vector field.

The vector field $\mathcal{A}^\mu$ is sometimes called the ‘torsion vector’ (see, e.g., ref. [\[34\]](#)).⁴ It is indeed directly related to the torsion of a connection compatible with the Galilean structure (see [\(3.11\)](#)). However, one could consider a torsion-free non-metric connection even in the case $\mathcal{A}^\mu \neq 0$. In this sense, it could be also be called the ‘non-metricity vector’. But because, regardless of the choice of connection, $\mathcal{A}^\mu$ encodes time dilation, we think that a more appropriate name is the *time dilation vector field*.

³In the literature, this condition is sometimes called the condition of *absolute simultaneity* or the *twistless-torsional Newton–Cartan* condition.

⁴In [\[34\]](#), the convention for unit timelike vector fields is sign-opposite to ours, i.e., $\tau_\alpha u^\alpha = -1$. Therefore, their definition of the torsion form is opposite to ours.

Finally, assuming the spacetime manifold is orientable and fixing an orientation, we can define the *spacetime volume form* of a Galilean structure. It may be defined geometrically as the unique 4-form η that takes the value 1 when evaluated on any positively oriented *Galilean basis* (e_t, e_1, e_2, e_3) , i.e., a positively oriented basis of the tangent space at p that is adapted to the Galilean structure in the sense of satisfying $\tau(e_t) = 1$ and $h|_p = \sum_{a=1}^3 e_a \otimes e_a$. In coordinate component notation, in positively oriented coordinates adapted to the spatial foliation, the spacetime volume form is given by

$$\eta_{\alpha\beta\mu\nu} := |\tau_0| (\det(h^{ij}))^{-1/2} \varepsilon_{\alpha\beta\mu\nu} , \quad (3.5)$$

where $\varepsilon_{\alpha\beta\mu\nu}$ is the antisymmetric Levi-Civita symbol, and $\det(h^{ij})$ is the determinant of the matrix of spatial components of $h^{\mu\nu}$. Given a choice of timelike vector field u^μ , the *spatial volume form* ${}^u\eta_{\alpha\mu\nu}$, which is annihilated by that vector field, is defined as

$${}^u\eta_{\alpha\mu\nu} := u^\beta \eta_{\beta\alpha\mu\nu} . \quad (3.6)$$

Remark 3.1. Any spacelike tensor can be pulled back uniquely to the spatial leaves. We call these leaves Σ , with the spacetime manifold taking the product form $\mathbb{R} \times \Sigma$. Equivalently, for any choice of timelike vector field u^μ , any tensor annihilated by it can be pulled back uniquely on Σ (see ref. [35]). In other words, spacelike tensors or tensors annihilated by u^μ are fully determined by their spatial components in a basis adapted to the spatial foliation. These spatial components define a unique tensor on each Σ . We will use this property to express the electric and magnetic Galilean Weyl tensors on Σ in [Section 5.2](#).

3.2. Galilean conformal structures and connections

In this section, we define Galilean conformal transformations, conformal Galilean structures, and their compatible connections (see ref. [18] and references therein for a detailed discussion of these definitions). These definitions will be important when assessing the conformal invariance of the Galilean Weyl tensor in [Section 5](#).

A *Galilean conformal transformation* of Galilean structures is defined as

$$\tau_\mu \xrightarrow{\varphi} e^\varphi \tau_\mu , \quad h^{\mu\nu} \xrightarrow{\varphi} e^{-2\varphi} h^{\mu\nu} , \quad (3.7)$$

where φ is a scalar field. We can extend this definition to any timelike vector field u^μ , its covariant spatial metric $\gamma_{\mu\nu}$, and its Coriolis field $\kappa_{\mu\nu}$ (see [Section 3.3](#) for its definition) as

$$u^\mu \xrightarrow{\varphi} e^{-\varphi} u^\mu , \quad \gamma_{\mu\nu} \xrightarrow{\varphi} e^{2\varphi} \gamma_{\mu\nu} , \quad \kappa_{\mu\nu} \xrightarrow{\varphi} e^\varphi \kappa_{\mu\nu} . \quad (3.8)$$

Note that, in general, Galilean conformal transformations (3.7) do not preserve closedness of the clock form, but preserve the condition $\tau \wedge d\tau = 0$. Since in the present paper we want to define a conformally invariant Galilean Weyl tensor, it is therefore essential to allow for non-closed, foliation-forming clock forms.

A *conformal Galilean structure*, denoted $[\tau, h]$, is an equivalence class of Galilean structures under Galilean conformal transformations. A connection ∇ is said to be *compatible* with a conformal Galilean structure $[\tau, h]$, or *conformally compatible*, if for some representative $(\tau_\mu, h^{\mu\nu})$ of the conformal structure there exists a 1-form n_μ such that

$$\nabla_\mu \tau_\nu = n_\mu \tau_\nu, \quad \nabla_\alpha h^{\mu\nu} = -2n_\alpha h^{\mu\nu}. \quad (3.9)$$

Under a Galilean conformal transformation of the representative, the 1-form n_μ transforms as $n_\mu \rightarrow n_\mu + \partial_\mu \varphi$.

3.3. Galilean structures and general affine connections

Given a Lorentzian metric $g_{\mu\nu}$, a general affine connection ∇ is uniquely determined by its non-metricity $\nabla_\alpha g_{\mu\nu}$ and its torsion $T^\alpha_{\mu\nu} = 2\Gamma^\alpha_{[\mu\nu]}$, which may be freely specified. The unique connection with vanishing torsion and non-metricity is the Levi-Civita connection of the metric. This gives rise, in particular, to the notions of *distortion*, *contortion* and *disformation*: the distortion is the difference of a general connection to the Levi-Civita one, the contortion is its torsion-dependent part, and the disformation is the non-metricity-dependent part.

A similar situation appears for Galilean structures. As shown in ref. [36], given a Galilean structure $(\tau_\mu, h^{\mu\nu})$ and a choice of timelike vector field u^μ , a general affine connection is uniquely determined by the non-metricities $Q_{\mu\nu} := \nabla_\mu \tau_\nu$ and $Q_\alpha^{\mu\nu} := \nabla_\alpha h^{\mu\nu}$, the torsion $T^\alpha_{\mu\nu} := 2\Gamma^\alpha_{[\mu\nu]}$, and the Coriolis (2-form) field with respect to u^μ defined as ${}^u\kappa_{\mu\nu} := (\nabla_{[\mu} u^\alpha) {}^u\gamma_{\nu]\alpha}$: it takes the form

$$\Gamma^\alpha_{\mu\nu} = {}^u\check{\Gamma}^\alpha_{\mu\nu} + 2\tau_{(\mu} \kappa_{\nu)}^\alpha + \gamma^\alpha_\beta Q_{(\mu\nu)}^\beta - \frac{1}{2}Q^\alpha_{\mu\nu} - u^\alpha Q_{(\mu\nu)} - T_{(\mu\nu)}^\alpha + \frac{1}{2}T^\alpha_{\mu\nu}, \quad (3.10a)$$

where indices have been lowered with ${}^u\gamma_{\mu\nu}$, we set

$${}^u\check{\Gamma}^\alpha_{\mu\nu} := u^\alpha \partial_{(\mu} \tau_{\nu)} + h^{\alpha\sigma} (\partial_{(\mu} \gamma_{\nu)\sigma} - \frac{1}{2} \partial_\sigma \gamma_{\mu\nu}), \quad (3.10b)$$

and where the non-metricities are constrained by

$$\tau_\sigma Q_\mu^{\nu\sigma} = -Q_\mu^{\nu\sigma}, \quad \tau_\sigma T^\sigma_{\mu\nu} = -2Q_{[\mu\nu]} + 2\omega_{\mu\nu}. \quad (3.11)$$

Conversely, the fields $Q_{\mu\nu}$, $Q_\alpha^{\mu\nu}$, $T^\alpha_{\mu\nu}$, $\kappa_{\mu\nu}$ may be freely specified, up to the constraints (3.11), and then the connection defined by (3.10) has these fields as its non-metricities, torsion, and Coriolis field with respect to u^μ .

In other words, $Q_\alpha^{\mu\nu}$, $Q_{\mu\nu}$, $T^\alpha_{\mu\nu}$ and $\kappa_{\mu\nu}$ are the ‘building blocks’ of any affine connection with respect to a Galilean structure, similarly to $\nabla_\alpha g_{\mu\nu}$ and $T^\alpha_{\mu\nu}$ being the building blocks with respect to a Lorentzian metric.

Fixing the connection, one can compute its Coriolis field with respect to a different timelike vector field $u^\mu - v^\mu$ (with v^μ spacelike): it is given by

$$\begin{aligned} {}^{u-v}\kappa_{\mu\nu} &= {}^u\kappa_{\mu\nu} - \partial_{[\mu} v_{\nu]} + \tau_{[\mu} \partial_{\nu]} \left(\frac{v^2}{2} \right) + \frac{1}{2} T^\rho_{\mu\nu} v_\rho \\ &\quad - Q_{[\mu\nu]\rho} v^\rho + \frac{1}{2} \tau_{[\mu} Q_{\nu]\rho\sigma} v^\rho v^\sigma + (u^\rho - v^\rho) (v_{[\mu} + v^2 \tau_{[\mu}) Q_{\nu]\rho}), \end{aligned} \quad (3.12)$$

where indices have been lowered with ${}^u\gamma_{\mu\nu}$, and $v^2 = v^\alpha v_\alpha$.

We see from (3.11) that, if $d\tau \neq 0$, any metric-compatible connection, i.e., one for which $Q_{\mu\nu} = 0$ and $Q_{\alpha}{}^{\mu\nu} = 0$, must feature torsion and, conversely, any torsion-free connection must feature non-metricities with respect to the Galilean structure. Additionally, even when $d\tau = 0$, differently to the Lorentzian case there is *not* a unique compatible torsion-free connection due to the freedom in the choice of Coriolis field. This is the main difference to the Lorentzian case. However, in the case $d\tau = 0$, given a choice of observer u^μ and of Coriolis field $\kappa_{\mu\nu}$, there is a unique compatible torsion-free connection, given by

$$\Gamma_{\mu\nu}^\alpha = \check{\Gamma}_{\mu\nu}^\alpha + 2\tau_{(\mu} \kappa_{\nu)}^\alpha . \quad (3.13)$$

Such torsion-free, metric-compatible connections are considered in classical Newton–Cartan gravity (see, e.g., refs. [7, 12]). However, once $d\tau \neq 0$, in addition to a Coriolis field, one needs a choice of torsion and non-metricities compatible with (3.11) to fix a connection. In the next section, we will discuss a condition that fixes a unique connection even in this case.

Remark 3.2. If the clock form is foliation forming, which we assume in the present paper, then there is always a unique *spatial* connection defined from the Galilean structure alone, namely the Levi-Civita connection of the induced Riemannian spatial metrics h_{ij} on the spatial leaves Σ .

3.4. A unique boost-invariant connection

The perspective on affine connections discussed in the previous section was a ‘classification’ one: *starting* with a connection ∇ , it determines—and is determined by—its torsion, non-metricities with respect to the Galilean structure, and Coriolis field with respect to a choice of timelike vector field u^μ , according to (3.10).

Here we now take a complementary perspective: we start with the Galilean structure, a unit timelike vector field u^μ , and a choice of Coriolis field, and want to *define* a connection from these ingredients—and possibly some additional structures—by some specified prescription. In this approach, one may in general choose the other ingredients to depend arbitrarily on u^μ , and hence the connections arising for different choices of u^μ could in general differ.

However, in discussions of physical theories based on Galilean geometry, one usually requires that the connection be *independent* of the choice of timelike vector field, since otherwise some preferred timelike vector field, or class of preferred timelike vector fields, could be defined (e.g., by requiring the connection to take some simple form). One demands that the theory should not possess a class of preferred observers, i.e., preferred timelike vector fields, from the geometric structure of the theory alone; any class of preferred observers should appear only once a field equation is considered.⁵ Therefore,

⁵For example, this is the case in standard Newton–Cartan gravity, for which the Newton–Cartan field equation induces a preferred class of observers called *Galilean observers* (see Definition 6.1).

because for our definition of Galilean Weyl tensors we want to stay at a geometric, ‘off-shell’ level, in the present paper we need to consider connections independent of the chosen timelike vector field.

In other words, we want to consider connections defined from $(\tau_\mu, h^{\mu\nu})$, a timelike vector field u^μ , and a Coriolis field in such a way that they are invariant under the transformation $u^\mu \rightarrow u^\mu - v^\mu$, where v^μ is a spacelike vector field. Such a transformation is called a (*local*) *Galilean boost* with parameter v^μ , and hence we call connections defined by such a prescription *boost-invariant connections*. Note that for a boost-invariant connection, the Coriolis field necessarily needs to transform under boosts according to (3.12): the connection is independent of the choice of u^μ , i.e., fixed, and hence the discussion from the previous section applies. The possibly remaining freedom in how to define a boost-invariant connection resides in the choice of torsion and non-metricities—which must also be boost-invariant (see [Proposition A.1](#) in [Appendix A](#) for details).

When $d\tau = 0$, there is a natural choice of torsion and non-metricities that yields, for a given choice of $(u^\mu, {}^u\kappa_{\mu\nu})$, a boost-invariant connection: we may choose the torsion and non-metricities to vanish, yielding the connection (3.13). In this case, the transformation law (3.12) for the Coriolis field simplifies to

$$\kappa_{\mu\nu} \xrightarrow{u \rightarrow u-v} \kappa_{\mu\nu} - \partial_{[\mu} \left(v_{\nu]} + \tau_{\nu]} \frac{v^2}{2} \right). \quad (3.14)$$

Interestingly, as we prove in [Appendix A](#), there is also a uniqueness result for a boost-invariant connection when we only require $\tau \wedge d\tau = 0$, similarly to the $d\tau = 0$ case:

Proposition 3.3. *Given a Galilean structure $(\tau_\mu, h^{\mu\nu})$ with $\tau \wedge d\tau = 0$ and a choice of timelike vector field and Coriolis field $(u^\mu, \kappa_{\mu\nu})$, there is a unique boost-invariant connection, denoted $\tilde{\nabla}$,*

- (i) *whose Coriolis field with respect to u^μ is the given $\kappa_{\mu\nu}$, and*
- (ii) *whose torsion and non-metricities are defined locally from $(\tau_\mu, h^{\mu\nu})$ and u^μ , depending on derivatives of $\tau_\mu, h^{\mu\nu}$ only to first order and linearly.*

It is given by

$$\tilde{\Gamma}_{\mu\nu}^\alpha = \check{\Gamma}_{\mu\nu}^\alpha + 2\tau_{(\mu} \kappa_{\nu)}^\alpha - \frac{1}{2} \gamma_{\mu\nu} \mathcal{A}^\alpha, \quad (3.15)$$

or, equivalently, by

$$\tilde{Q}_{\mu\nu} = \omega_{\mu\nu}, \quad \tilde{Q}_\alpha{}^{\mu\nu} = -\delta_\alpha^{(\mu} \mathcal{A}^{\nu)}, \quad \tilde{T}^\alpha{}_{\mu\nu} = 0. \quad (3.16)$$

Under boosts, the Coriolis field transforms as

$$\kappa_{\mu\nu} \xrightarrow{u \rightarrow u-v} \kappa_{\mu\nu} - \partial_{[\mu} \left(v_{\nu]} + \tau_{\nu]} \frac{v^2}{2} \right). \quad (3.17)$$

Remark 3.4. When $d\tau = 0$, the connection $\tilde{\nabla}$ reduces to the classical connection (3.13).

To our knowledge, this is the first time this connection is derived, as boost-invariant connections considered in the literature are always constructed with extra structures on top of $(\tau_\mu, h^{\mu\nu}, u^\mu, \kappa_{\mu\nu})$, such as a mass gauge field (see ref. [34] and references therein). We include a presentation and discussion of these approaches in [Appendix B](#).

The uniqueness of the connection (3.15), along with how its Coriolis field transforms under Galilean boosts, is reminiscent of the natural connection (3.13) obtained from a closed clock form. In this sense, the connection (3.15) is the most natural connection that should be considered from $(\tau_\mu, h^{\mu\nu}, u^\mu, \kappa_{\mu\nu})$ when $\tau \wedge d\tau = 0$.

4. The Galilean limit

The definition of the Galilean Weyl tensor which we will provide in [Section 5](#) will be obtained from the Galilean limit of the Lorentzian Weyl tensor. We detail that limit in the present section. In particular, we provide formulae for a non-closed clock form, and we provide the limit of the 1+3 quantities. As already remarked in [Section 1.1](#), we use the terminology ‘Galilean’ rather than ‘Newtonian’ for denoting the $c \rightarrow \infty$ limit of Lorentzian geometry to highlight the fact that this is a geometric (off-shell) approach (see also our discussion in [Section 6.3](#) on the difference between a ‘Galilean theory’ and a ‘Newtonian theory’).

4.1. The ansatz

In the classical ansatz for the Galilean limit, the first two leading orders of the Lorentzian metric have the form

$$\hat{g}^{\mu\nu} = h^{\mu\nu} + \lambda (-\hat{u}^\mu \hat{u}^\mu + k^{\mu\nu}) + \mathcal{O}(\lambda^2), \quad (4.1a)$$

$$\hat{g}_{\mu\nu} = -\frac{1}{\lambda} \tau_\mu \tau_\nu + \hat{\gamma}_{\mu\nu} - 2\phi \tau_\mu \tau_\nu + \mathcal{O}(\lambda^1), \quad (4.1b)$$

where $\hat{u}^\mu$ is a timelike vector field for which we write $\hat{\gamma}_{\mu\nu} := \hat{u}^\alpha \gamma_{\mu\nu}$, ϕ is a scalar field, and $k^{\mu\nu}$ is a spacelike tensor.

Remark 4.1. In the geometry arising from the Galilean limit, there is a gauge freedom corresponding to the application of λ -dependent diffeomorphisms to the Lorentzian objects [37, 34, 11]. In particular, this includes the gauge transformation $\hat{u}^\mu \rightarrow \hat{u}^\mu - \psi \mathcal{A}^\mu - h^{\mu\alpha} \partial_\alpha \psi$, where ψ is a scalar field. Interestingly, this gauge freedom leaves unchanged the vorticity tensor of any observer (see [Section 4.2.2](#), in particular [Remark 4.3](#)).

For a general Galilean limit, the 1-form τ_μ is not necessarily exact, nor foliation forming. In the literature, it is however almost always assumed to be foliation forming, i.e., to satisfy $\tau \wedge d\tau = 0$. This can be justified from the limit of the Einstein equation with some assumptions on the leading order of the energy-momentum tensor [38, 39, 34]. A stronger result was obtained in ref. [33], where the authors showed that, prior to

considering any field equations, the Frobenius integrability condition is necessarily satisfied if global hyperbolicity is assumed for the Lorentzian structure, and that the relation $\tau = e^\psi dt$ holds globally, where ψ and t are scalar fields. We assume $\tau \wedge d\tau = 0$ for the rest of the paper.

Remark 4.2. Ergen et al. [40] argued that choosing $\tau = dt$ is a gauge choice, without loss of generality for the limit, corresponding to a choice of foliation with respect to which performing the limit. However, as we will be interested in conformal transformations of the Galilean Weyl tensor, we cannot fix the clock form to be exact. Furthermore, one can argue that even though the choice of foliation might be seen as a choice of gauge on the Lorentzian level, performing the limit with respect to different foliations corresponds to taking the point of view of different families of observers, which are (in the limit) not related by Galilean boost transformations—i.e., when ‘forgetting’ the Lorentzian origin of the Galilean theory, the choice of foliation becomes a *physical* one.

The spacetime volume form $\epsilon_{\alpha\beta\mu\nu}^\lambda$ of the metric $g_{\mu\nu}^\lambda$ becomes

$$\epsilon_{\alpha\beta\mu\nu}^\lambda := \sqrt{\lambda \left| \det(g_{\sigma\omega}^\lambda) \right|} \epsilon_{\alpha\beta\mu\nu} = \eta_{\alpha\beta\mu\nu} + \mathcal{O}(\lambda^1). \quad (4.2)$$

The LO term of the Levi-Civita connection is $\Gamma_{\mu\nu}^{\alpha(-1)}$, and we have

$$\Gamma_{\mu\nu}^{\alpha(-1)} = -\tau_\mu \tau_\nu \mathcal{A}^\alpha, \quad (4.3)$$

$$\Gamma_{\mu\nu}^{\alpha(0)} = \hat{u}^\alpha \tilde{\Gamma}_{\mu\nu}^\alpha + \tau_\mu \tau_\nu (h^{\alpha\sigma} \partial_\sigma \phi - 2\phi \mathcal{A}^\alpha - k^{\alpha\sigma} \mathcal{A}_\sigma) - \hat{u}^\alpha \tau_{(\mu} \hat{\mathcal{A}}_{\nu)}, \quad (4.4)$$

where we write $\hat{\mathcal{A}}_\mu := \hat{\gamma}_{\mu\alpha} \mathcal{A}^\alpha$.

We see that the zeroth order term of the Levi-Civita connection is a connection of the form (A.10) with $c_1 = c_2 = c_4 = c_5 = 0$, $c_3 = -1$, $d_i = 0$, and where the Coriolis field with respect to $\hat{u}^\mu$ is $\hat{\kappa}_{\mu\nu} := \hat{u}^\alpha \kappa_{\mu\nu} = \tau_{[\mu} (\hat{\gamma}^{\alpha}_{\nu]} \partial_\alpha \phi - 2\phi \hat{\mathcal{A}}_{\nu]} - \hat{\gamma}_{\nu]\alpha} k^{\alpha\beta} \hat{\mathcal{A}}_\beta)$. Therefore, this connection is not boost-invariant, as it is different from the unique boost-invariant connection (3.15) (with the same Coriolis field $\hat{\kappa}_{\mu\nu}$). However, when the clock form is closed, then $\Gamma_{\mu\nu}^{\alpha(-1)} = 0$, and $\Gamma_{\mu\nu}^{\alpha(0)}$ corresponds to the unique torsion-free compatible connection, with $\hat{\kappa}_{\mu\nu} = \tau_{[\mu} \hat{\gamma}^{\alpha}_{\nu]} \partial_\alpha \phi$.

The zeroth order connection is related to the boost-invariant connection $\tilde{\nabla}$ (with the same Coriolis field $\hat{\kappa}_{\mu\nu}$) by

$$\Gamma_{\mu\nu}^{\alpha(0)} = \tilde{\Gamma}_{\mu\nu}^\alpha + \frac{1}{2} \hat{\gamma}_{\mu\nu} \mathcal{A}^\alpha - \tau_{(\mu} \hat{\mathcal{A}}_{\nu)} \hat{u}^\alpha. \quad (4.5)$$

Note that the spatial projection of the Coriolis field vanishes, i.e., we have

$$\hat{\kappa}^{\mu\nu} = h^{\alpha[\mu} \hat{\gamma}^{\nu]}_{\beta} \tilde{\nabla}_\alpha \hat{u}^\beta = 0. \quad (4.6)$$

The LO term of the Riemann tensor is $R^{(-1)}_{\beta\mu\nu}$, and we have

$$R^{(-1)}_{\beta\mu\nu} = 2\tau_\beta\tau_{[\mu} \left(\tilde{\nabla}_{\nu]} \mathcal{A}^\alpha - \frac{1}{2} \hat{\mathcal{A}}_{\nu]} \mathcal{A}^\alpha \right), \quad (4.7a)$$

$$R^{(0)}_{\beta\mu\nu} = \tilde{\nabla} R^{(0)}_{\beta\mu\nu} + \tau_\beta\tau_{[\mu} \left(\hat{u}^\alpha \mathcal{A}^\sigma \mathcal{L}_{\hat{u}} \hat{\gamma}_{\nu]\sigma} - 2\mathcal{A}^\alpha \partial_{\nu]} \phi \right) - \mathcal{A}^\alpha \tau_{[\mu} \mathcal{L}_{\hat{u}} \hat{\gamma}_{\nu]\beta}, \quad (4.7b)$$

and for the Ricci tensor

$$R^{(-1)}_{\mu\nu} = -\tau_\mu\tau_\nu \tilde{\nabla}_\alpha \mathcal{A}^\alpha, \quad (4.8a)$$

$$R^{(0)}_{\mu\nu} = \tilde{\nabla} R^{(0)}_{\mu\nu} + \tau_\mu\tau_\nu \hat{u}^\sigma \partial_\sigma \phi + \tau_{(\mu} \mathcal{L}_{\hat{u}} \hat{\gamma}_{\nu)\sigma} \mathcal{A}^\sigma, \quad (4.8b)$$

where $\tilde{\nabla} R^{(0)}_{\beta\mu\nu}$ is the Riemann tensor of the connection (4.4) arising as the zeroth order term of the Levi-Civita connection. Note that we used the unique boost-invariant connection $\tilde{\nabla}$ in these expressions.

4.2. Limit of the kinematical quantities

In this section, we describe the Galilean limit of the kinematical quantities. We will use these later when expressing the electric and magnetic Galilean Weyl tensors spatially.

4.2.1. Limit of U -orthogonal quantities

Any vector field $\hat{U}^\mu$ timelike with respect to the Lorentzian metric has an expansion of the form $\hat{U}^\mu = u^\mu + \mathcal{O}(\lambda^1)$ where u^μ is unit timelike with respect to the Galilean structure obtained in the limit. Furthermore, for the associated one-form we have $\hat{U}_\mu = -\frac{1}{\lambda} \tau_\mu + \mathcal{O}(\lambda^0)$, and

$$\hat{U}_{b\mu\nu} = {}^u\gamma_{\mu\nu} + \mathcal{O}(\lambda^1), \quad \hat{U}^{b\mu\nu} = h^{\mu\nu} + \mathcal{O}(\lambda^1), \quad \hat{U}^{b\mu}{}_\nu = {}^u\gamma^\mu{}_\nu + \mathcal{O}(\lambda^1). \quad (4.9)$$

Therefore, as for the timelike vector field, the Lorentzian orthogonal projector at LO corresponds to the Galilean covariant spatial metric for the (0,2) version, to the Galilean spatial projector for the (1,1) version, and to the spatial metric of the Galilean structure for the (2,0) version. There are two main consequences of this fact:

- (i) For any timelike vector field $\hat{U}^\mu$, any contravariant tensor (say, p^μ) orthogonal to $\hat{U}^\mu$ will be annihilated by τ_μ at LO, and therefore $\hat{p}^{(0)\mu}$ defines, after pull-back, a spacelike tensor on Σ (see [Remark 3.1](#)). Using a basis adapted to the spatial foliation, we can denote that tensor with spatial indices as $\hat{p}^{(0)i}$. Therefore, while before taking the limit, a tensor orthogonal to $\hat{U}^\mu$ cannot be defined on a ‘spatial foliation’ if $\hat{U}^\mu \Omega_{\mu\nu} \neq 0$ (since no such foliation exists), at LO that tensor always defines uniquely a spacelike tensor on the Galilean spatial foliation.

- (ii) At LO, the spatial ‘connection’ $\overset{\lambda}{U}D_\mu$ with respect to $\overset{\lambda}{U}b_{\mu\nu}$ (defined in (2.10)) corresponds, after pull-back to Σ , to the unique spatial connection (denoted D_i) associated to the Galilean structure, namely the Levi-Civita connection of the Riemannian metric induced on Σ by $h^{\mu\nu}$. The same holds for the LO of the ‘Riemann tensor’ of that ‘connection’, whose pull-back to Σ we denote by $\mathcal{R}_{ij}$. Therefore, at LO, $\overset{\lambda}{U}D_\mu$ has all the properties of an affine connection on Σ , even if $\overset{\lambda}{U}\Omega_{\mu\nu} \neq 0$ (see Remark 2.1).

4.2.2. Galilean kinematical quantities

Given a Lorentzian timelike vector field $\overset{\lambda}{U}^\mu$ with LO term u^μ , the LO terms of the kinematical quantities are

$$\overset{\lambda}{U}\Theta^{\mu\nu} = h^{\alpha\mu}h^{\beta\nu} \frac{1}{2}\mathcal{L}_u {}^u\gamma_{\alpha\beta} + \mathcal{O}(\lambda^1), \quad (4.10a)$$

$$\overset{\lambda}{U}\Omega^{\mu\nu} = h^{\alpha\mu}h^{\beta\nu} \partial_{[\alpha}w_{\beta]} + \mathcal{A}^{[\mu}w^{\nu]} + \mathcal{O}(\lambda^1), \quad (4.10b)$$

$$\overset{\lambda}{U}a^\mu = -\frac{1}{\lambda}\mathcal{A}^\mu + \mathcal{O}(\lambda^1), \quad (4.10c)$$

where $w^\mu := u^\mu - \hat{u}^\mu$ is the spacelike ‘tilt’ / boost vector field between the considered timelike vector field u^μ and the one defined by the Galilean limit (4.1), and the index on w_β was lowered with ${}^u\gamma_{\mu\nu}$. We want to define the Galilean expansion and vorticity tensors of u^μ as these LO fields. For the (Galilean) expansion tensor of u^μ , this yields the definition

$${}^u\Theta^{\mu\nu} := h^{\alpha\mu}h^{\beta\nu} \frac{1}{2}\mathcal{L}_u \gamma_{\alpha\beta}. \quad (4.11)$$

Note that due to $\gamma_{\alpha\beta}u^\alpha = 0$, the Lie derivative $\mathcal{L}_u \gamma_{\alpha\beta}$ vanishes when contracted with u^α (or u^β), such that the index-lowered expansion tensor takes the simple form ${}^u\Theta_{\mu\nu} = \frac{1}{2}\mathcal{L}_u \gamma_{\mu\nu}$. For the (Galilean) vorticity tensor of u^μ , we obtain the definition

$${}^u\Omega^{\mu\nu} := h^{\alpha\mu}h^{\beta\nu} \partial_{[\alpha}w_{\beta]} + \mathcal{A}^{[\mu}w^{\nu]}. \quad (4.12)$$

Note that this expression depends not only on u^μ and the Galilean structure $(\tau_\mu, h^{\mu\nu})$, but also on the timelike vector field $\hat{u}^\mu$ arising in the Galilean limit, since $w^\mu = u^\mu - \hat{u}^\mu$. What are we to make of this dependence? By definition, the Galilean vorticity of $\hat{u}^\mu$ vanishes, i.e., $\hat{u}\Omega^{\mu\nu} = 0$. Further, if $\check{u}^\mu$ is any timelike vector field with vanishing Galilean vorticity, we see directly that the vorticity of u^μ can be written as

$${}^u\Omega^{\mu\nu} = h^{\alpha\mu}h^{\beta\nu} \partial_{[\alpha}\check{w}_{\beta]} + \mathcal{A}^{[\mu}\check{w}^{\nu]} \quad (4.13)$$

with $\check{w}^\mu = u^\mu - \check{u}^\mu$. In this sense, the notion of Galilean vorticity depends, in addition to the Galilean structure, on the family of vorticity-free observers (timelike vector fields), which is fixed by the Galilean limit, as it depends on the tilt $\check{w}_\mu$ with respect to any of these observers. Note that this dependence is already present in the classical case $d\tau = 0$.

By rewriting the first term of (4.13) in terms of the boost-invariant connection $\tilde{\nabla}$ arising in the Galilean limit, the dependence on the family of vorticity-free observers can in a certain sense be completely ‘pushed’ into the second, $\mathcal{A}^\mu$ -dependent term, as follows. By a direct computation, one can show

$$h^{\alpha\mu}h^{\beta\nu}\partial_{[\alpha}\tilde{\omega}_{\beta]} = h^{\alpha[\mu}\gamma^{\nu]\beta}\tilde{\nabla}_\alpha u^\beta - h^{\alpha[\mu}\tilde{\gamma}^{\nu]\beta}\tilde{\nabla}_\alpha \tilde{u}^\beta, \quad (4.14)$$

where we wrote $\tilde{\gamma}^\mu_\nu := \tilde{u}^\mu \gamma^\mu_\nu$. This is precisely the difference of the spatial projections of the Coriolis fields of $\tilde{\nabla}$ with respect to u^μ and $\tilde{u}^\mu$. Thus, the vorticity can be written as

$${}^u\Omega^{\mu\nu} = {}^u\kappa^{\mu\nu} - \tilde{u}^\mu \kappa^{\mu\nu} + \mathcal{A}^{[\mu}\tilde{\omega}^{\nu]}, \quad (4.15)$$

where, as above, $\tilde{u}^\mu$ is a member of the family of vorticity-free timelike vector fields. In particular, in terms of the field $\hat{u}^\mu$ arising in the Galilean limit, by (4.6) we have

$${}^u\Omega^{\mu\nu} = {}^u\kappa^{\mu\nu} + \mathcal{A}^{[\mu}\hat{u}^{\nu]}. \quad (4.16)$$

In the classical case $d\tau = 0$, the vorticity of u^μ can hence be rewritten in a form that does not depend *explicitly* on the family of vorticity-free observers, at the expense of expressing it in terms of the spacetime connection. Nevertheless, even in this case ${}^u\Omega_{ij}$ is still imposed to be an exact (spatial) 2-form by (4.14), which is related, in particular, to the so-called *Newtonian condition* introduced by Trautman [3, 4].

Note that the Galilean expansion and vorticity tensors as defined here reduce to those from standard Newton–Cartan gravity when assuming $d\tau = 0$ (see [12, Section 3.2 and Remark 3.51]).

Remark 4.3. In the classical Galilean limit, i.e., with $d\tau = 0$, the family of vorticity-free observers is easily characterised: any two of them differ by a spacelike vector field whose associated spatial one-form is (spatially) closed. In what follows, we generalise this result in the presence of a non-trivial time dilation vector field.

Taking the exterior derivative of the equation $d\tau = \tau \wedge \mathcal{A}$, we obtain $0 = -\tau \wedge d\mathcal{A}$, implying that the spatial pull-back $\mathcal{A}_i$ of the time dilation vector field to any spatial leaf Σ is a closed (spatial) 1-form. In particular, locally we have $\mathcal{A}_i = -\partial_i \Psi$ for some scalar field Ψ , i.e.,

$$\mathcal{A}|_\Sigma = -d(\Psi|_\Sigma). \quad (4.17)$$

Let now $\tilde{\omega}^\mu$ be the difference between an arbitrary timelike vector field u^μ and a vorticity-free timelike vector field $\tilde{u}^\mu$. From (4.13), we know that u^μ is itself vorticity-free if and only if the spacelike form $\tilde{\omega}_i$ satisfies

$$0 = d(\tilde{\omega}|_\Sigma) + \mathcal{A} \wedge \tilde{\omega}. \quad (4.18)$$

Combined with (4.17), this is equivalent to

$$0 = d(\tilde{\omega}|_\Sigma) - d(\Psi|_\Sigma) \wedge \tilde{\omega} = e^\Psi d(e^{-\Psi}\tilde{\omega}|_\Sigma), \quad (4.19)$$

i.e., to

$$\tilde{\omega}_i = e^\Psi n_i, \quad (4.20)$$

where n_i is a spatially closed 1-form, $d(n|_\Sigma) = 0$.

In other words, the Galilean limit implies the existence of a family of (Galilean) vorticity-free observers that all differ by a spatial velocity that is locally of the form (4.20). This form includes in particular the gauge freedom on $\hat{u}^\mu$ discussed in Remark 4.1, for which n_i is exact.

As mentioned in Section 4.1, global hyperbolicity of the Lorentzian structure imposes $\tau = e^\Psi dt$ to hold globally, such that (4.17) holds globally, and the difference of vorticity-free observers has the form (4.20) globally.

The Galilean expansion and vorticity tensors (4.11), (4.12) are spacelike with respect to the Galilean structure, and therefore define tensors on each spatial leaf Σ (see Section 4.2.1). We denote these spatial tensors by ${}^u\Theta_{ij}$ and ${}^u\Omega_{ij}$, respectively, and call them by the names of (Galilean) *expansion* and *vorticity* tensors of u^μ as well. We will discuss some of their properties in Section 6.2.

Let us stress again that, while the LO term of the spatial connection ${}^\lambda\hat{D}_\mu$ does not depend on the choice of timelike vector field, the LO terms of the expansion and vorticity tensors *do* depend on that vector field—they are properties of the vector field, not of the spacetime. For two different Galilean timelike vector fields u^μ and m^μ , differing by the spacelike vector field $v^\mu := u^\mu - m^\mu$, from (4.11) and (4.12), their Galilean expansion and vorticity tensors differ by

$${}^m\Theta_{ij} - {}^u\Theta_{ij} = -D_{(i}v_{j)} , \quad {}^m\Omega_{ij} - {}^u\Omega_{ij} = -\partial_{[i}v_{j]} - \mathcal{A}_{[i}v_{j]} . \quad (4.21)$$

Finally, to later assess the conformal invariance of the Galilean Weyl tensor, we need to determine how the Galilean kinematical quantities change under conformal transformation. From (2.20) for the transformation of the Lorentzian kinematical quantities, as well as the transformation of the spatial connection and spatial Ricci tensor, we obtain

$$\theta \xrightarrow{\varphi} e^{-\varphi} (\theta + 3 \mathcal{L}_u \varphi) , \quad \sigma_{ij} \xrightarrow{\varphi} e^\varphi \sigma_{ij} , \quad \Omega_{ij} \xrightarrow{\varphi} e^\varphi \Omega_{ij} , \quad \mathcal{A}_i \xrightarrow{\varphi} \mathcal{A}_i - D_i \varphi , \quad (4.22a)$$

$${}^h\Gamma_{ij}^k \xrightarrow{\varphi} {}^h\Gamma_{ij}^k + 2h^k_{(i}D_{j)}\varphi - h_{ij}D^k\varphi , \quad (4.22b)$$

$$\mathcal{R}_{ij} \xrightarrow{\varphi} \mathcal{R}_{ij} - D_i D_j \varphi + D_i \varphi D_j \varphi - \left(\Delta \varphi + D_k \varphi D^k \varphi \right) h_{ij} , \quad (4.22c)$$

where ${}^h\Gamma_{ij}^k$ are the Levi-Civita coefficients of the spatial metric.

5. The Galilean Weyl tensor

In this section we show how to define a Galilean Weyl tensor that is manifestly conformally invariant. We first provide a definition involving a spacetime connection, then we provide a definition involving the spatial connection and the kinematical quantities. Both approaches are off-shell.

5.1. Spacetime approach

Here, we will discuss how to give a definition of a Galilean Weyl tensor using a spacetime connection by consideration of the Galilean limit.

5.1.1. Definition

To obtain a definition of the Galilean Weyl tensors (i.e., versions of ‘the’ Galilean Weyl tensor with different index placement), we consider the Galilean limit of the corresponding Lorentzian versions, and keep only the LO terms.

For simplicity, in the spacetime approach, we consider only a timelike vector field $\hat{U}^\mu$ such that its LO term is $\hat{u}^\mu$, i.e., the timelike vector field naturally defined by the Galilean limit (4.1). In other words, we consider a timelike vector field $\hat{U}^\mu$ for which the Galilean vorticity vanishes (see Section 4.2.2). Only in the spatial approach of Section 5.2 will a general timelike vector field be considered.

The LO terms of the electric and magnetic parts of the Lorentzian Weyl tensor with respect to $\hat{U}^\mu$, defined by (2.2) and (2.3), are

$$\begin{aligned}\hat{E}_{\mu\nu} &:= E_{\mu\nu}^{(-1)} = \frac{1}{2} \hat{\gamma}^\alpha{}_{\langle\mu} \hat{\gamma}^\beta{}_{\nu\rangle} R_{\alpha\beta}^{(0)} - \hat{\gamma}^\alpha{}_{\langle\mu} \hat{\gamma}^\beta{}_{\nu\rangle} \tilde{\nabla}_\alpha \hat{\mathcal{A}}_\beta + \hat{\mathcal{A}}_{\langle\mu} \hat{\mathcal{A}}_{\nu\rangle} , \\ &= \frac{1}{2} \hat{\gamma}^\alpha{}_{\langle\mu} \hat{\gamma}^\beta{}_{\nu\rangle} \nabla R_{\alpha\beta}^{(0)} - \hat{\gamma}^\alpha{}_{\langle\mu} \hat{\gamma}^\beta{}_{\nu\rangle} \tilde{\nabla}_\alpha \hat{\mathcal{A}}_\beta + \hat{\mathcal{A}}_{\langle\mu} \hat{\mathcal{A}}_{\nu\rangle} ,\end{aligned}\quad (5.1)$$

$$\begin{aligned}\hat{H}_{\mu\nu} &:= H_{\mu\nu}^{(0)} = \frac{1}{2} \eta_{\alpha\beta(\mu} h^{\beta\lambda} R_{\lambda\nu)\sigma}^{(0)} \hat{u}^\sigma \\ &= \frac{1}{2} \eta_{\alpha\beta(\mu} h^{\beta\lambda} \nabla R_{\lambda\nu)\sigma}^{(0)} \hat{u}^\sigma + \frac{1}{4} \mathcal{A}^\alpha h^{\sigma\beta} \eta_{\alpha\sigma(\mu} \mathcal{L}_{\hat{u}} \hat{\gamma}_{\nu)\beta} .\end{aligned}\quad (5.2)$$

Using (2.4), we can express the LO term of the full Weyl tensor in terms of the LO terms of the electric and magnetic parts. Considering the different ‘versions’ of the full Weyl tensor, we obtain

$$\hat{C}_{\alpha\beta\mu\nu}^\lambda = \frac{1}{\lambda} \left[-2\tau_\alpha \tau_{[\mu} \hat{E}_{\nu]\beta} + 2\tau_\beta \tau_{[\mu} \hat{E}_{\nu]\alpha} \right] + \mathcal{O}(\lambda^0), \quad (5.3a)$$

$$\hat{C}^\lambda_{\beta\mu\nu} = \frac{1}{\lambda} \left[-2\tau_\beta \tau_{[\mu} \hat{E}_{\nu]}^\alpha \right] + \mathcal{O}(\lambda^0), \quad (5.3b)$$

$$\hat{C}^{\alpha\beta}{}_{\mu\nu} = 4 \left(\hat{\gamma}^{[\alpha}{}_{[\mu} \hat{E}^{\beta]}{}_{\nu]} - \hat{u}^{[\alpha} \tau_{[\mu} \hat{E}^{\beta]}{}_{\nu]} \right) - 2\eta^{\alpha\beta\sigma} \tau_{[\mu} \hat{H}_{\nu]\sigma} + \mathcal{O}(\lambda^1), \quad (5.3c)$$

$$\hat{C}^{\alpha\beta\mu}{}_\nu = 2 \left(h^{\mu[\alpha} \hat{E}^{\beta]}{}_\nu - \hat{\gamma}^{[\alpha}{}_\nu \hat{E}^{\beta]\mu} + \tau_\nu \hat{u}^{[\alpha} \hat{E}^{\beta]\mu} \right) - \tau_\nu \eta^{\alpha\beta\sigma} h^{\mu\omega} \hat{H}_{\sigma\omega} + \mathcal{O}(\lambda^1), \quad (5.3d)$$

$$\hat{C}^{\alpha\beta\mu\nu} = 2 \left(h^{\alpha[\mu} \hat{E}^{\nu]\beta} - h^{\beta[\mu} \hat{E}^{\nu]\alpha} \right) + \mathcal{O}(\lambda^1), \quad (5.3e)$$

where we recall that indices on the Galilean objects are raised with $h^{\mu\nu}$.

We see that the LO terms of the (0,4), the (1,3), and the (4,0) versions of the Weyl tensor depend only on the LO term of the electric part, while the LO terms of the (2,2) and the

(3,1) versions depend on the LO terms of both the electric and magnetic parts. In this sense, the (2,2) and the (3,1) Weyl tensors carry more geometric information than the other versions and for this reason they are more suited to define a Galilean Weyl tensor. This peculiarity does not occur for the electric and magnetic parts themselves, for which the LO terms carry the same information regardless of the limit being taken with upper or lower indices. In this sense, *what carries all the information on the Weyl tensor at LO are its electric and magnetic parts.*

Note also that for the different versions of the Weyl tensor, the LO is *different*: the (0,4) and the (1,3) versions have a non-vanishing $\frac{1}{\lambda}$ coefficient, while the others start at order λ^0 . (In this sense, the (0,4) and (1,3) Weyl tensors diverge in the Galilean $c \rightarrow \infty$ limit.)

There is, however, an issue in the LO expressions given so far: the Ricci and Riemann tensors involved in these formulae are not defined with respect to a boost-invariant connection, but with respect to the zeroth order connection (4.4). It is therefore preferable to express them in terms of the unique boost-invariant connection (3.15). Using (4.5) to compute $\overset{(0)}{\nabla} R^\alpha_{\beta\mu\nu} - \tilde{R}^\alpha_{\beta\mu\nu}$, we obtain

$$\hat{E}_{\mu\nu} = \frac{1}{2} \hat{\gamma}^\alpha_{\langle\mu} \hat{\gamma}^\beta_{\nu\rangle} \tilde{R}_{\alpha\beta} - \hat{\gamma}^\alpha_{\langle\mu} \hat{\gamma}^\beta_{\nu\rangle} \tilde{\nabla}_\alpha \hat{\mathcal{A}}_\beta + \frac{3}{4} \hat{\mathcal{A}}_{\langle\mu} \hat{\mathcal{A}}_{\nu\rangle} , \quad (5.4)$$

$$\hat{H}_{\mu\nu} = \frac{1}{2} \eta_{\alpha\beta} \tilde{R}^\alpha_{\beta\mu\nu} \hat{u}^\sigma + \frac{1}{4} \mathcal{A}^\alpha \eta_{\alpha\beta} \mathcal{L}_{\hat{u}} \hat{\gamma}_{\nu\rangle\beta} . \quad (5.5)$$

We can now give a definition for the off-shell Galilean Weyl tensor, its electric part, and its magnetic part.

Definition 5.1 (Galilean Weyl tensor, spacetime version). *Let $(\tau_\mu, h^{\mu\nu})$ be a Galilean structure, $(u^\mu, \kappa_{\mu\nu})$ be a timelike vector field and a choice of spatially vanishing Coriolis field, i.e $\kappa^{\mu\nu} = 0$, and $\tilde{\nabla}$ be the unique boost-invariant connection (3.15) constructed from $(\tau_\mu, h^{\mu\nu}, u^\mu, \kappa_{\mu\nu})$. The (off-shell) Galilean electric and magnetic Weyl tensor parts with respect to u^μ are defined, respectively, as*

$$E_{\mu\nu} := \frac{1}{2} \gamma^\alpha_{\langle\mu} \gamma^\beta_{\nu\rangle} \tilde{R}_{\alpha\beta} - \gamma^\alpha_{\langle\mu} \gamma^\beta_{\nu\rangle} \tilde{\nabla}_\alpha \mathcal{A}_\beta + \frac{3}{4} \mathcal{A}_{\langle\mu} \mathcal{A}_{\nu\rangle} , \quad (5.6)$$

$$H_{\mu\nu} := \frac{1}{2} \eta_{\alpha\beta} h^{\beta\lambda} \tilde{R}^\alpha_{\lambda\mu\nu} u^\sigma + \frac{1}{4} \mathcal{A}_\alpha \eta^{\alpha\beta} \mathcal{L}_u \gamma_{\nu\rangle\beta} . \quad (5.7)$$

The (2,2) Galilean Weyl tensor is defined as

$$C^{\alpha\beta}_{\mu\nu} := 4 \left(\gamma^{[\alpha}_{[\mu} E^{\beta]}_{\nu]} - u^{[\alpha} \tau_{[\mu} E^{\beta]}_{\nu]} \right) - 2 \eta^{\alpha\beta\sigma} \tau_{[\mu} H_{\nu]\sigma} . \quad (5.8)$$

Proposition 5.2. *The Galilean Weyl tensor $C^{\alpha\beta}_{\mu\nu}$, its electric part $E_{\mu\nu}$, and its magnetic part $H_{\mu\nu}$ are invariant under Galilean conformal transformations.*

Proof. Because the Lorentzian electric and magnetic parts $\overset{\lambda}{U} E_{\mu\nu}$ and $\overset{\lambda}{U} H_{\mu\nu}$ are invariant under Lorentzian conformal transformations, and because the Galilean limit of a Lorentzian conformal transformation is a Galilean conformal transformation, their LO terms are automatically invariant under Galilean conformal transformations. $\square$

Note that when $d\tau = 0$, the definition (5.8) reduces to

$$C^{\alpha\beta}{}_{\mu\nu} = h^{\sigma\beta} R^{\alpha}{}_{\sigma\mu\nu} - 2\delta^{\alpha}{}_{[\mu} R_{\nu]\sigma} h^{\beta]\sigma} + \frac{1}{3}\delta^{\alpha}{}_{[\mu} \delta^{\beta}{}_{\nu]} R_{\sigma\omega} h^{\sigma\omega} . \quad (5.9)$$

Because all information on the Galilean Weyl tensor is contained in the electric and magnetic parts, and because there is no unique way to define a ‘full Galilean Weyl tensor’ from the electric and magnetic parts, for the rest of this paper we will only discuss these two parts, and not anymore consider the full object (5.8).

5.1.2. Discussion

It is important to note that the precise formulae for the electric and magnetic parts depend on which spacetime connection we choose to express the Ricci and Riemann tensors. We chose the connection derived in Section 3.4 because we wanted a boost-invariant connection that does not require the introduction of an additional structure. As discussed in Appendix B, by introducing an extra structure, e.g., a mass gauge field M_μ , other boost-invariant connections could be considered, changing essentially the last terms in the formulae (5.6) and (5.7), while changing nothing in the formula (5.8) for the (2,2) Galilean Weyl tensor. In any case, regardless of the connection that is chosen, the formulae will always be conformally invariant and equivalent.

For instance, by introducing the mass gauge field M_μ and the boost-invariant fields $\bar{u}^\mu$, $\bar{u}\gamma_{\mu\nu}$, and $\bar{\mathcal{A}}_\mu$ (see Appendix B.2 for the definitions), we can choose the connection to be either *torsionful and compatible*, taking the form

$$\bar{\Gamma}_{\mu\nu}^\alpha = \bar{u}\check{\Gamma}_{\mu\nu}^\alpha + 2\tau_{(\mu} \bar{u}\kappa_{\nu)}^\alpha + \bar{u}\gamma_{(\mu}^\alpha \bar{\mathcal{A}}_{\nu)} - \bar{u}\gamma_{\mu\nu} \bar{\mathcal{A}}^\alpha + \bar{u}\gamma_{[\mu}^\alpha \bar{\mathcal{A}}_{\nu]} + \bar{u}\gamma_{[\mu}^\alpha \tau_{\nu]} \bar{\mathcal{A}}_{\nu]} , \quad (5.10)$$

or *torsion-free and conformally compatible* (see definition (3.9)), taking the form

$$\bar{\Gamma}_{\mu\nu}^\alpha = \bar{u}\check{\Gamma}_{\mu\nu}^\alpha + 2\tau_{(\mu} \bar{u}\kappa_{\nu)}^\alpha + 2\bar{u}\gamma_{(\mu}^\alpha \bar{\mathcal{A}}_{\nu)} - \bar{u}\gamma_{\mu\nu} \bar{\mathcal{A}}^\alpha + \bar{u}^\alpha \tau_{(\mu} \bar{\mathcal{A}}_{\nu)} . \quad (5.11)$$

In these two cases, the electric and magnetic parts become simply

$$E_{\mu\nu} = \frac{1}{2} \bar{u}\gamma_{(\mu}^\alpha \bar{u}\gamma_{\nu)}^\beta \bar{R}_{\alpha\beta} , \quad H_{\mu\nu} = \frac{1}{2} \eta_{\alpha\beta(\mu} h^{\beta\lambda} \bar{R}^\alpha{}_{\lambda\nu)\sigma} \bar{u}^\sigma . \quad (5.12)$$

These formulae are equivalent to those from Definition 5.1, but written with respect to another spacetime connection. They have the merit of being simpler, but at the cost of introducing extra structure.

Another clear disadvantage of Definition 5.1 is that the objects which appear therein are difficult to interpret, regardless of the spacetime connection we choose. In particular, it is not clear how the spatial curvature can be obtained from the spacetime connection, unless $d\tau = 0$. For this reason, a more enlightening way of defining the electric and magnetic parts of the Galilean Weyl tensor would be to use only spacelike quantities, like the spatial curvature and kinematical quantities. The clear advantage of such a ‘spatial

approach' is that there is a unique spatial connection that can be constructed from the Galilean structure, namely the Levi-Civita connection of the (pull-back) Riemannian metric h_{ij} on Σ . Therefore, there will be a unique way of representing the electric and magnetic parts. We pursue this approach in the next section.

Finally, note that **Definition 5.1** defines the electric and magnetic parts with respect to an irrotational observer, as imposed by the condition $\kappa^{\mu\nu} = 0$. In principle, it would be possible to obtain a definition for a general observer by taking the limit with respect to a timelike vector field $\hat{U}$ with a LO term different from $\hat{u}$, which would result in more complicated formulae. We will provide such a general definition only in the spatial approach in the next section.

Note also that when lowering the second index of the $d\tau = 0$ form (5.9) of the Galilean Weyl tensor with $\gamma_{\mu\nu}$, the resulting object fulfills all the conditions demanded in ref. [29] of a Galilean Weyl tensor, apart from spatial flatness.

5.2. Spatial approach

Here, we will discuss a 'purely spatial' definition of a Galilean Weyl tensor.

5.2.1. Definition

From the expressions (2.17) and (2.18) of the electric and magnetic parts of the Lorentzian Weyl tensor in terms of the kinematical quantities, and the limit of kinematical quantities presented in Section 4.2, we obtain the following definition:

Definition 5.3 (Galilean Weyl tensor, spatial version). *Let $(\tau_\mu, h^{\mu\nu})$ be a Galilean structure such that τ_μ is foliation forming with leaves Σ and with time dilation vector field $\mathcal{A}^\mu$. Given a timelike vector field u^μ with Galilean shear $\sigma_{\mu\nu}$ and vorticity $\Omega_{\mu\nu}$, the electric and magnetic parts of the Galilean Weyl tensor on Σ with respect to u^μ are defined as*

$$E_{ij} := \frac{1}{2} \left(\mathcal{R}_{\langle ij \rangle} - D_{\langle i} \mathcal{A}_{j \rangle} + \mathcal{A}_{\langle i} \mathcal{A}_{j \rangle} \right), \quad (5.13)$$

$$H_{ij} := \eta_{ab\langle i} \left[D^a \left(\sigma^b{}_{j \rangle} + \Omega^b{}_{j \rangle} \right) + \mathcal{A}_{j \rangle} \Omega^{ab} \right], \quad (5.14)$$

where D_i and $\mathcal{R}_{ij}$ are, respectively, the Levi-Civita connection of the spatial metric h_{ij} and its Ricci tensor.

Note that in the spatial approach, we have only defined the electric and magnetic parts of the Galilean Weyl tensor, not any 'spatial Galilean Weyl tensor' as a combined object.

By a direct computation, one can show:

Proposition 5.4. *The (spatial) electric part E_{ij} and magnetic part H_{ij} of the Galilean Weyl tensor are invariant under Galilean conformal transformations (4.22).*

Because D_i , $\mathcal{R}_{ij}$ and $\mathcal{A}_i$ are independent of the choice of observer, the electric part is independent as well. This is not the case for the magnetic part.

5.2.2. Relation to the spacetime approach

The spatial [Definition 5.3](#) of the electric and magnetic parts of the Galilean Weyl tensor is equivalent to the (pull-back of the) spacetime [Definition 5.1](#), because they both correspond to the LO terms of the electric and magnetic parts of the Lorentzian Weyl tensor. However, while there was the freedom of the choice of spacetime connection in [Definition 5.1](#), there is no such freedom in [Definition 5.3](#).

To obtain the spacetime [Definition 5.1](#) from the spatial [Definition 5.3](#) without consideration of the Galilean limit, one should express the spatial covariant derivative and the spatial Ricci tensor in terms of the chosen spacetime connection. When $d\tau = 0$, that relation is similar to the relativistic one ([2.10](#)) (see, e.g., [[35](#)]): the spatial Levi-Civita connection is simply the restriction of the spacetime connection. However, this is not the case when $d\tau \neq 0$. For instance, restriction of the unique boost-invariant connection $\tilde{\nabla}_\mu$ does not lead to a connection D_i that is compatible with the spatial Riemannian metric. A more elaborate definition is required, which we do not aim to study in this paper as we already established the equivalence between [Definition 5.1](#) and [Definition 5.3](#).

The fact that, when $d\tau \neq 0$, there arises an ambiguity in the spacetime connection that should be considered in the Galilean limit, and no such ambiguity arises for the spatial connection, may be taken as an indication that, in the $d\tau \neq 0$ case, to understand the geometry and the physics of a Galilean invariant theory it is easier *not* to work with a spacetime representation *à la* standard Newton–Cartan geometry, but to work instead with a spatial representation, closer to the classical Newtonian formulation.

6. Discussion

The approach we followed in [Section 5](#) for the definition of a Galilean Weyl tensor—or its electric and magnetic parts—is completely off-shell. In particular, we did not impose the spatial curvature to be zero, contrary to what has been done so far in the literature using Galilean structures to define the Galilean Weyl tensor [[28–30](#)], and the literature using a more classical approach to the Newtonian limit [[41–43](#)]. This has two advantages. First, since spatial curvature is not preserved under conformal transformations, having non-zero spatial curvature is the only way in which one can define a conformally invariant Galilean Weyl tensor. Second, the Weyl tensor of Lorentzian structures is a geometric quantity defined independently of any field equations. It is therefore natural to require that a Galilean version should also be defined geometrically.

Let us elaborate on the first point above. One might wonder how it can be, given what we have just said, that Dewar and Read [[30](#)] profess to show that their *on-shell* Galilean Weyl tensor is invariant under Galilean conformal transformations. The key to resolving this puzzle is to note that the conformal rescalings chosen in ref. [[30](#)] are *not* the same as the conformal rescalings used in this article (or in refs. [[18, 44](#)]): in fact, those scalings suffice to make the conformal transformations considered in that article *projective* transformations (on which see, e.g., ref. [[10](#)]). And since the ‘Galilean Weyl

tensor' considered in refs. [28–30] is in fact what is known as the *projective* Weyl tensor (see, e.g., ref. [45]), which is a *projective* invariant, it is of course no surprise that that object is invariant under those alternative transformations.

This, in turn, leads to a related puzzle: how can it be the case that, by taking the Galilean limit of a relativistic Weyl tensor which is *conformally* invariant, refs. [28, 29] obtain an object which is *projectively* invariant? The answer to this question (setting aside the fact that there are some mathematical slips in ref. [28] when it comes to taking the limit) is that it is known from ref. [45] that, in the relativistic setting, the projective and conformal Weyl tensors agree when one is working with an Einstein manifold (for which, by definition, $g_{\mu\nu} \propto R_{\mu\nu}$). But since when one goes on-shell in Newton–Cartan theory one effectively has this condition (recall the Newton–Cartan field equation $R_{\mu\nu} = 4\pi G\rho\tau_\mu\tau_\nu$), it is little surprise that one obtains a projective invariant in the end. Of course, if one follows the approach of the present paper and remains fully off-shell, what one arrives at for the Galilean Weyl tensor (as in Definition 5.8) is *not* a projective invariant (but *is* invariant under the ‘correct’ Galilean conformal transformations⁶).

In any case, let us return now to our off-shell definitions as presented above. From the discussion in the previous section, in order to understand the Galilean Weyl tensors, it is sufficient to consider only the electric and magnetic parts, since the full spacetime Galilean Weyl tensor (5.8) is uniquely given in terms of these two parts. (Of course, in the spatial definition, we have only defined these parts separately.)

6.1. Interpretation of the electric part

As a consequence of our approach, the usual interpretation of the electric part of the Weyl tensor as corresponding to the tidal tensor fails for our Galilean version. Indeed, our definition corresponds to the LO term of the Lorentzian electric part, which is an order-(−1) off-shell object, while the tidal tensor only appears at zeroth order, which agrees with the LO term only on-shell, i.e., assuming the Newton–Cartan equation. Being the (−1)-order term, our definition depends directly on the traceless part of the spatial Ricci tensor. Therefore, to obtain the usual relation between the electric part and the tidal tensor, a second electric Galilean Weyl tensor would need to be introduced, corresponding to the next-to-leading-order. We do not provide such a definition in the present paper.

6.2. Interpretation of the magnetic part

Contrary to the electric part, the off-shell magnetic part of the Weyl tensor depends on the choice of observer, as it depends explicitly on the shear and vorticity tensors. Additionally, as for the on-shell approach (with the Newton–Cartan equation), it is the zeroth order term of the Lorentzian magnetic part. The classical result is that the magnetic part vanishes in Newtonian gravity. This result holds on-shell with the Newton–Cartan equation. As we show below, off-shell this is not the case in general.

⁶For a more general approach to Galilean conformal transformations which subsumes both choices of scaling, see ref. [17]; this more general definition is also used in ref. [46].

As shown in [Section 4.2.2](#), the Galilean limit imposes that there exists a vector field w_i such that any vorticity tensor has the form

$$\Omega_{ij} = \partial_{[i} w_{j]} + \mathcal{A}_{[i} w_{j]} , \quad (6.1)$$

and the difference between the vorticities of two different observers (see [\(4.21\)](#)) implies that there exist irrotational observers.

Contrary to the vorticity tensor, no constraint on the expansion tensor arises in the Galilean limit. In particular, it is not guaranteed that there exists an observer with vanishing expansion tensor, in particular for cosmological situations. As a consequence, the magnetic part does not vanish in general. To understand in which situation it does vanish, it is useful to consider the scalar–vector–tensor (svt) decomposition of symmetric tensors. It states that, in 3 dimensions, given a Riemannian metric h_{ij} and a symmetric tensor field A_{ij} , there exist a unique scalar field χ , vector field gradient $D_{(i} v_{j)}$, and transverse-traceless (tt) tensor field X_{ij} such that $A_{ij} = \chi h_{ij} + D_{(i} v_{j)} + X_{ij}$. The existence and uniqueness of this decomposition require either a closed manifold, or specific asymptotic conditions at infinity (e.g., asymptotic flatness) [\[47, 48\]](#). The former corresponds to a cosmological setup, the latter to an isolated system. We consider one of these two situations in the following, such that the svt decomposition applies.

We apply the svt decomposition to the expansion tensor Θ_{ij} , writing it as

$$\Theta_{ij} = \chi h_{ij} + D_{(i} v_{j)} + X_{ij} . \quad (6.2)$$

Inserting [\(6.1\)](#) and [\(6.2\)](#) into the definition [\(5.14\)](#) of the magnetic part of the Galilean Weyl tensor, one may show that it takes the form

$$\begin{aligned} H_{ij} = & \frac{1}{2} \eta^{ab} ({}_i \mathcal{R}_{j)a} (v_b + w_b) + \eta^{ab} {}_{\langle i} \left[\frac{1}{2} D_a D_{j \rangle} (v_b - w_b) + D_a X_{j \rangle b} \right. \\ & \left. + \frac{1}{2} D_a (\mathcal{A}_b w_{j \rangle}) - \frac{1}{2} D_a (w_b \mathcal{A}_{j \rangle}) + \mathcal{A}_{j \rangle} (D_a w_b + \mathcal{A}_a w_b) \right] . \end{aligned} \quad (6.3)$$

Note that in this equation, the traceless part is only taken on the indices i and j , and not on the enclosed indices a, b ‘in between’. In order to derive [\(6.3\)](#), one needs to express the commutator of spatial covariant derivatives in terms of the spatial Riemann tensor, and then express the Riemann tensor (which is possible in 3 dimensions) in terms of the metric and the Ricci tensor.

Again, in general, without any assumptions on the kinematical quantities, the magnetic part [\(6.3\)](#) does not vanish. However, when $\mathcal{A}_i = 0$ and the Newton–Cartan equation is assumed, then $\mathcal{R}_{ij} = 0$, and for any observer we have [\[35\]](#)⁷

$$D_i \chi = 0 , \quad D_i v_j = D_i w_j , \quad \Delta X_{ij} = 0 . \quad (6.4)$$

The last condition can be derived from $\frac{1}{2} \partial_t h_{ij} = \Theta_{ij}$ and $\partial_t \mathcal{R}_{ij} = 0$, where ∂_t is the time derivative along the observer (i.e., timelike vector field) with expansion tensor Θ_{ij} .

⁷This is a consequence of the mixed spatial-temporal projection of the Newton–Cartan equation.

Using the boundary conditions needed for the svT decomposition theorem to hold, the last condition in (6.4) implies $0 = \int_{\Sigma} dx^3 \sqrt{h} X^{ij} \Delta X_{ij} = - \int_{\Sigma} dx^3 \sqrt{h} D^k X^{ij} D_k X_{ij}$. Then, using $D^k X^{ij} D_k X_{ij} \geq 0$, we obtain $D_k X_{ij} = 0$. Consequently, X_{ij} is curl-free, i.e., $D_{[a} X_{b]j} = 0$. Therefore (6.4) implies that the magnetic part vanishes for any observer.

This result, obtained in the context of the Newton–Cartan equation, suggests the following definition of a specific type of observers that can also be considered off-shell, which we call *Galilean observers*:

Definition 6.1 (Galilean observers). *An observer is called Galilean if its expansion and vorticity tensors satisfy*

$$\Theta_{ij} = H(t)h_{ij} + X_{ij} , \quad \Omega_{ij} = 0 , \quad (6.5)$$

where X_{ij} is a curl-free $\tau\tau$ tensor. In other words, an observer is Galilean if it is vorticity-free and the svT decomposition of its expansion tensor has no gradient part and a curl-free $\tau\tau$ part.

Remark 6.2. In the standard approach to the Galilean limit, i.e., with $\mathcal{A}_i = 0$, it is sometimes assumed that the decompositions (6.1) and (6.2) of the vorticity and expansion tensors feature the same vector field, i.e., $v_i = w_i$, and that we have $\Theta_{ij} = D_{(i} v_{j)}$ and $\Omega_{ij} = D_{[i} v_{j]}$ (see, e.g., [41, 42]). As discussed above, this assumption is not generally true, for two reasons: (i) the Galilean limit alone does not constrain the expansion tensor and its relation to the vorticity tensor, and (ii) even assuming the Newton–Cartan equation, the two additional terms $H(t)h_{ij}$ and X_{ij} are, in general, present in the expansion tensor, representing the total volume expansion of the spatial manifold (if finite) and the anisotropic expansion, respectively. If the spatial manifold is $\mathbb{R}^3$, i.e., assuming asymptotic flatness for the svT decomposition theorem to hold, then we necessarily have $H(t) = 0$ and $X_{ij} = 0$.

The existence of Galilean observers is a key implication of the Newton–Cartan equation. In particular, when $X_{ij} = 0$, it ensures that there exists a coordinate system adapted to the spatial foliation, and such that the spatial metric is $h_{ij} = a(t)^2 \bar{h}_{ij}(x^k)$, where $\dot{a}/a := \chi(t)$ and $\bar{h}_{ij}(x^k)$ is time-independent. This is not guaranteed without the existence of Galilean observers. Additionally, when the Newton–Cartan equation is considered, the *opposite* to the 4-acceleration of these observers (with respect to a compatible torsion-free connection) is a solution of the (cosmological) Poisson equation, and therefore corresponds to the Newtonian gravitational field [35]. For these reasons, these observers play a fundamental role in Newtonian gravitation.

By the definition of Galilean observers (and (6.3)), we have:

Proposition 6.3. *The magnetic Galilean Weyl tensor with respect to any Galilean observer vanishes.*

Therefore, if one wants the magnetic part to vanish, then, rather than considering a specific field equation, a natural (pre-field-equation, i.e., geometric/kinematical) assumption is to rather assume the existence of Galilean observers. If we do so, then, for any observer, their expansion and vorticity tensors satisfy

$$\Theta_{ij} = H(t)h_{ij} + D_{(i} v_{j)} + X_{ij} , \quad \Omega_{ij} = \partial_{[i} v_{j]} + \mathcal{A}_{[i} v_{j]} , \quad (6.6)$$

where X_{ij} is a curl-free $\tau\tau$ tensor, and the magnetic part of the Weyl tensor is

$$H_{ij} = \eta_{ab} \langle i \mathcal{R}_{j \rangle a} v_b + \eta^{ab} \langle i \left[\frac{1}{2} D_a (\mathcal{A}_b v_{j \rangle}) - \frac{1}{2} D_a (v_b \mathcal{A}_{j \rangle}) + \mathcal{A}_{j \rangle} (D_a v_b + \mathcal{A}_a v_b) \right] \rangle. \quad (6.7)$$

Taking $v^i = 0$ corresponds to considering a Galilean observer, and we get $H_{ij} = 0$.

6.3. Condition for a ‘Galilean’ theory to be a ‘Newtonian’ theory

When considering the Newton–Cartan equation alone, without deriving the Galilean structure from a Galilean limit, the existence of irrotational (i.e., vorticity-free) observers is not guaranteed. For this reason, Trautman introduced a condition on connections in standard Newton–Cartan gravity guaranteeing the existence of irrotational observers [3, 4]; later, connections satisfying this condition were dubbed ‘Newtonian’ connections [6]. The condition may, for example, be written as the curvature condition $R^\mu{}_\nu{}^\rho{}_\sigma = R^\rho{}_\sigma{}^\mu{}_\nu$ (see, e.g., [12, Section 2.4]).⁸ This lead to the distinction between ‘Galilean geometries/theories’ and ‘Newtonian geometries/theories’: both are constructed from Galilean structures, but only the latter fulfils the Newtonian condition. More generally, a Galilean theory is considered physically relevant if it is Newtonian.

Inspired by our discussion in the previous section, and in the spirit of this classical ‘Newtonian’ condition introduced by Trautman, it seems natural to require that a Galilean theory should be called ‘Newtonian’ only if there exist (Galilean) observers with vanishing magnetic part of the Galilean Weyl tensor. The status of this additional condition for the ‘Newtonianity’ of a theory is, however, not the same as Trautman’s ‘Newtonian’ condition: the latter follows automatically in the off-shell Galilean limit from the interchange symmetry of the Lorentzian Riemann tensor, while the former condition is not imposed by the Galilean limit, and in this sense is added by hand.

6.4. Galilean structures with spatial curvature

Since the off-shell electric part of the Galilean Weyl tensor features the spatial curvature, it is worth mentioning situations where such a curvature appears in Galilean theories. There are two main situations:

- (i) One may consider Galilean theories with spatial curvature to reproduce, at LO, effects that are usually obtained at post-Newtonian orders in the classical (spatially flat) approach, like the perihelion precession of Mercury (see, e.g., refs. [50, 51, 34]). In this approach, the spatial curvature is sourced by the time dilation vector field. This approach does not aim at describing new physics, but rather at reproducing known one from the classical limit. In this sense, the presence of spatial curvature is a choice.

⁸Actually, even when this condition is considered, only the existence observers with a harmonic vorticity, i.e., $\Delta\Omega_{ij} = 0$, are guaranteed (see [49] and references therein). Therefore, deriving the Galilean structure from a Galilean limit (with $\tau \wedge d\tau = 0$) is really what guarantees the existence of irrotational observers.

- (ii) Spatial curvature also arises when the spatial manifold has a topology not allowing for a Euclidean, i.e., flat, metric [7, 52–55], e.g., with $\Sigma \cong \mathbb{S}^3$ or $\Sigma \cong \mathbb{S} \times \mathbb{S}^2$. In this situation, the presence of spatial curvature is not a choice (contrary to the previous example), but a necessity imposed by topology. It was first shown by Künzle [7, Remark on p. 448]⁹, and later more systematically studied in refs. [53–55], that, to get a physically consistent Newtonian theory with spatial curvature (of topological origin), the Newton–Cartan equation should feature a background Ricci curvature,

$$R_{\mu\nu} - \bar{R}_{\mu\nu} = 4\pi G\rho\tau_\mu\tau_\nu, \quad (6.8)$$

where $\bar{R}_{\mu\nu}$ is constructed from the canonical (or topological) Ricci curvature of the considered topology. That canonical curvature is given by the Thurston classification of 3-dimensional topological spaces.

As discussed in Section 6.3, in both of these cases, for the Galilean theory to be physically relevant, or in other words ‘Newtonian’, the existence of Galilean observers should be demanded. The presence of these observers implies some conditions on the time evolution of the spatial curvature. Indeed, taking $\mathcal{G}_t$ to be the time derivative along a Galilean observer, i.e., $\frac{1}{2}\mathcal{G}_th_{ij} = H(t)h_{ij} + X_{ij}$, we obtain

$$\begin{aligned} \mathcal{G}_t\mathcal{R}_{ij} &= \Delta_L \left(\frac{1}{2}\mathcal{G}_th_{ij} \right) - D_i D_j \left(\frac{1}{2}h^{kl}\mathcal{G}_th_{kl} \right) \\ &= \Delta_L X_{ij} = \Delta X_{ij} = 3\mathcal{R}_{k(i}X_{j)}^k - h_{ij}\mathcal{R}_{kl}X^{kl} - \frac{1}{2}\mathcal{R}X_{ij}, \end{aligned} \quad (6.9)$$

where Δ_L is the Lichnerowicz Laplacian.¹⁰ Conversely, (6.9) constrains the anisotropic expansion tensor X_{ij} if the spatial curvature is prescribed.

7. Conclusion

In this paper, we have provided manifestly conformally invariant off-shell definitions for the Galilean Weyl tensor, its electric part, and its magnetic part. We followed two approaches: a Galilean Weyl tensor can be defined (i) by using a spacetime connection (Definition 5.1), or (ii) by using the spatial connection and kinematical quantities (Definition 5.3). In both approaches, the clock form was foliation forming but not exact.

While the former approach is closer to the usual Newton–Cartan formalism in which equations are formulated in terms of the Galilean structure, a spacetime connection, and its curvature tensor, the freedom in the choice of that connection (due to $d\tau \neq 0$) makes the definitions of the Galilean Weyl tensors not unique, and difficult to interpret. In Definition 5.1, we chose the unique boost-invariant connection (3.15) that can be derived from a Galilean structure and a choice of timelike vector field and Coriolis field. But, as

⁹See also the early work of Zöllner [56], which is discussed in ref. [57].

¹⁰The Lichnerowicz Laplacian is defined as $\Delta_L Y_{ij} := -\Delta Y_{ij} + 2D^k D_{(i} Y_{j)k}$ for divergence-free, symmetric Y_{ij} . If in addition Y_{ij} is curl-free, one has $\Delta_L Y_{ij} = \Delta Y_{ij}$ without a minus sign, as used in (6.9) applied to X_{ij} . The last equality in (6.9) follows by a standard computation using that X_{ij} is divergence- and curl-free, and expressing the 3-dimensional Riemann tensor in terms of the metric and the Ricci tensor.

discussed in [Section 5.1.2](#), other connections could be considered, and therefore, this remains a choice.

The purely spatial definition ([Definition 5.3](#)) of the electric and magnetic parts of the Galilean Weyl tensor does not suffer from the issues of the spacetime approach because there is a unique spatial connection to consider, namely the Levi-Civita connection of the spatial metric. In this approach, the electric and magnetic parts are expressed in terms of the kinematical quantities of an observer, making these definitions much more geometrically transparent. Contrary to the on-shell approach, the electric part is not related to the tidal tensor but to the spatial curvature, and the magnetic part does not vanish in general.

In the spirit of the ‘Newtonian’ condition for standard Newton–Cartan gravity, introduced by Trautman, we argued that requiring the existence of specific observers for which the magnetic part vanishes, called *Galilean observers*, should be a necessary condition for a Galilean theory to be considered physical, hence called ‘Newtonian’. We argued that this condition should be considered when constructing any Galilean invariant theory featuring spatial curvature.

Stepping back, our work helps to clarify and contextualise existing results for the Galilean Weyl tensors presented in refs. [\[28–30\]](#), and to understand their limitations when they are considered off-shell. But—even more importantly—it should pave the way for future directions of research, for example:

- (i) With a fully general understanding of Galilean Weyl tensors in hand, it is natural to ask whether this object can be leveraged in ways analogous to the relativistic context—for example, in developing a non-relativistic analogue of the Petrov classification of spacetimes, or in understanding non-relativistic gravitational radiation (on which see ref. [\[58\]](#)).
- (ii) One can ask whether properties of the relativistic Weyl tensor carry over to the Galilean regime. For example, does a vanishing Galilean Weyl tensor imply conformal flatness of the Galilean structure?
- (iii) Since there has been much recent work on Galilean conformal field theory [\[59, 60\]](#) and twistor theory [\[61, 62, 17\]](#), it is natural to ask whether the Galilean Weyl tensor will constitute an object of study in those contexts going forward.

Acknowledgements

P.K.S. and Q.V. thank Pembroke College, Oxford, for hospitality during a stay where the present research was conducted. Q.V. was supported in part by the European Research Council (ERC) under the European Union’s Horizon 2020 research and innovation program (grant agreement ERC advanced grant 740021-ARTHUS, principal investigator Thomas Buchert). For helpful discussions J.R. thanks Niels Linnemann, Eleanor March, and the audience of a talk at UC Irvine.

A. Proof of uniqueness of the boost-invariant connection

Here we prove the uniqueness result for the boost-invariant connection discussed in [Section 3.4](#).

First, we prove a very helpful statement:

Proposition A.1. *Let $(\tau_\mu, h^{\mu\nu})$ be a Galilean structure. A connection is boost-invariant if and only if its non-metricities with respect to $(\tau_\mu, h^{\mu\nu})$ and torsion are boost-invariant and its Coriolis field transforms under boosts according to [\(3.12\)](#), i.e., satisfies*

$$\begin{aligned} \kappa_{\mu\nu} \xrightarrow{u \rightarrow u-v} & \kappa_{\mu\nu} - \partial_{[\mu} v_{\nu]} + \tau_{[\mu} \partial_{\nu]} \left(\frac{v^2}{2} \right) + \frac{1}{2} T^\rho{}_{\mu\nu} v_\rho \\ & - Q_{[\mu\nu]\rho} v^\rho + \frac{1}{2} \tau_{[\mu} Q_{\nu]\rho\sigma} v^\rho v^\sigma + (u^\rho - v^\rho)(v_{[\mu} + v^2 \tau_{[\mu} Q_{\nu]\rho}) \end{aligned} \quad (\text{A.1})$$

Proof. The proof of this statement is essentially an exercise in precisely keeping track of which objects depend on a choice of (unit) timelike vector field (with respect to $(\tau_\mu, h^{\mu\nu})$). We will carry this out in a very explicit way, such as to be as clear as possible.

For the sake of explicitness, for the period of this proof we introduce the notation

$$\kappa[u; \nabla]_{\mu\nu} := (\text{Coriolis field of } \nabla \text{ with respect to } u)_{\mu\nu} = (\nabla_{[\mu} u^{\alpha]} u^{\gamma} \gamma_{\nu]\alpha}) \quad (\text{A.2})$$

for any affine connection ∇ and timelike vector field u^μ . According to the discussion from [Section 3.3](#), the Coriolis fields of a fixed connection with respect to different timelike vector fields satisfy

$$\begin{aligned} \kappa[u - v; \nabla]_{\mu\nu} &= \kappa[u; \nabla]_{\mu\nu} - \partial_{[\mu} v_{\nu]} + \tau_{[\mu} \partial_{\nu]} \left(\frac{v^2}{2} \right) + \frac{1}{2} T^\rho{}_{\mu\nu} v_\rho \\ & - Q_{[\mu\nu]\rho} v^\rho + \frac{1}{2} \tau_{[\mu} Q_{\nu]\rho\sigma} v^\rho v^\sigma + (u^\rho - v^\rho)(v_{[\mu} + v^2 \tau_{[\mu} Q_{\nu]\rho}) \end{aligned} \quad (\text{A.3})$$

for u^μ a timelike and v^μ a spacelike vector field, where $Q_\alpha{}^{\mu\nu} = \nabla_\alpha h^{\mu\nu}$, $Q_{\mu\nu} = \nabla_\mu \tau_\nu$ are the non-metricities and $T^\alpha{}_{\mu\nu}$ is the torsion of ∇ .

Now we are going to more precisely formulate the statement we want to prove. Explicitly stated in fully written-out form, it means the following: for a map $u \mapsto \nabla[u]$ from the set of timelike vector fields to the set of affine connections, we have

$$\nabla[u] = \nabla[\tilde{u}] \text{ for all timelike vector fields } u^\mu, \tilde{u}^\mu \quad (\text{A.4})$$

(i.e., the map is constant) if and only if we have

$$\nabla[u]_\mu \tau_\nu = \nabla[\tilde{u}]_\mu \tau_\nu, \quad \nabla[u]_\alpha h^{\mu\nu} = \nabla[\tilde{u}]_\alpha h^{\mu\nu}, \quad T[\nabla[u]]^\alpha{}_{\mu\nu} = T[\nabla[\tilde{u}]]^\alpha{}_{\mu\nu} \quad (\text{A.5a})$$

for all timelike vector fields $u^\mu, \tilde{u}^\mu$ and

$$\begin{aligned} \kappa[u - v; \nabla[u - v]]_{\mu\nu} &= \kappa[u; \nabla[u]]_{\mu\nu} - \partial_{[\mu} v_{\nu]} + \tau_{[\mu} \partial_{\nu]} \left(\frac{v^2}{2} \right) + \frac{1}{2} T^\rho{}_{\mu\nu} v_\rho \\ & - Q_{[\mu\nu]\rho} v^\rho + \frac{1}{2} \tau_{[\mu} Q_{\nu]\rho\sigma} v^\rho v^\sigma + (u^\rho - v^\rho)(v_{[\mu} + v^2 \tau_{[\mu} Q_{\nu]\rho}) \end{aligned} \quad (\text{A.5b})$$

for any timelike vector field u^μ and spacelike vector field v^μ , where the non-metricities and torsion are independent of u^μ by [\(A.5a\)](#).

Now we are going to prove the claimed equivalence of (A.4) and (A.5). First assuming that the connection is boost-invariant, i.e., $\nabla[u] = \nabla$ is a fixed connection, its non-metricities and torsion are independent of u , i.e., we have (A.5a); further, the equation (A.3) relating the Coriolis fields of a fixed connection with respect to different timelike vector fields directly translates into (A.5b).

For the converse, assume that the equations (A.5) hold. We fix a timelike vector field u^μ and a spacelike vector field v^μ , and consider the two connections $\nabla[u]$ and $\nabla[u - v]$. By (A.5a), we know that these have agreeing non-metricities and torsion. Further, combining the general equation (A.3) applied to $\nabla = \nabla[u - v]$ with the assumed equation (A.5b) yields

$$\kappa[u; \nabla[u - v]]_{\mu\nu} = \kappa[u; \nabla[u]]_{\mu\nu} . \quad (\text{A.6})$$

Therefore, the two connections $\nabla[u]$ and $\nabla[u - v]$ have agreeing non-metricities, torsion, and Coriolis field with respect to u . Therefore, according to the classification theorem [36] they agree, $\nabla[u] = \nabla[u - v]$. Thus we have shown that $\nabla[u]$ is independent of u , finishing the proof. $\square$

Now we can prove uniqueness and existence of the boost-invariant connection discussed in Section 3.4.

Proof of Proposition 3.3. Let $(\tau_\mu, h^{\mu\nu})$ be a Galilean structure with $\tau \wedge d\tau = 0$, u^μ a timelike vector field, and $\kappa_{\mu\nu}$ an arbitrary two-form.

We consider an affine connection ∇ for which we fix $(\nabla_{[\mu} u^{\sigma]} \gamma_{\nu]\sigma} = \kappa_{\mu\nu}$, and whose torsion and non-metricities are defined locally from $(\tau_\mu, h^{\mu\nu})$ and u^μ , depending on derivatives of $\tau_\mu, h^{\mu\nu}$ only to first order and linearly. Consequently, the torsion and non-metricities are tensor fields depending pointwise on $(\tau_\mu, h^{\mu\nu})$ and u^μ , and pointwise linearly on $(d\tau)_{\mu\nu}$ —i.e., on $\mathcal{A}^\mu$ as we assume $\tau \wedge d\tau = 0$ —and on $\mathcal{L}_u h^{\mu\nu}$. However, regarding $\mathcal{L}_u h^{\mu\nu}$, only its spatial part $-\frac{1}{2} \gamma_{\mu\alpha} \gamma_{\nu\beta} \mathcal{L}_u h^{\alpha\beta} = \frac{1}{2} \mathcal{L}_u \gamma_{\mu\nu} = {}^u\Theta_{\mu\nu}$ is independent of $\mathcal{A}^\mu$.¹¹

Grouping together the torsion and non-metricity terms, we therefore consider a connection of the form

$$\Gamma_{\mu\nu}^\alpha = {}^u\check{\Gamma}_{\mu\nu}^\alpha + 2\tau_{(\mu} \kappa_{\nu)}^\alpha + C^\alpha{}_{\mu\nu}[\mathcal{A}^\mu, \Theta_{\mu\nu}] , \quad (\text{A.7})$$

where $\check{\Gamma}_{\mu\nu}^\alpha$ is defined in (3.10b) and $C^\alpha{}_{\mu\nu}[\mathcal{A}^\mu, \Theta_{\mu\nu}]$ is a (1,2)-tensor depending pointwise on $(\tau_\mu, h^{\mu\nu})$, on u^μ , and linearly on $\mathcal{A}^\mu$ and $\mathcal{L}_u \gamma_{\mu\nu}$. Therefore, the only possible independent terms in $C^\alpha{}_{\mu\nu}$ are

$$\begin{aligned} & \gamma^\alpha{}_{(\mu} \mathcal{A}_{\nu)} , & \gamma_{\mu\nu} \mathcal{A}^\alpha , & u^\alpha \tau_{(\mu} \mathcal{A}_{\nu)} , & \tau_\mu \tau_\nu \mathcal{A}^\alpha , & \gamma^\alpha{}_{[\mu} \mathcal{A}_{\nu]} , & u^\alpha \tau_{[\mu} \mathcal{A}_{\nu]} , \\ & \gamma^\alpha{}_{(\mu} \tau_{\nu)} \theta , & \gamma_{\mu\nu} u^\alpha \theta , & u^\alpha \tau_\mu \tau_\nu \theta , & \gamma^\alpha{}_{[\mu} \tau_{\nu]} \theta , & & \\ & \Theta^\alpha{}_{(\mu} \tau_{\nu)} , & \Theta_{\mu\nu} u^\alpha , & \Theta^\alpha{}_{[\mu} \tau_{\nu]} , & & & \end{aligned} \quad (\text{A.8})$$

¹¹We have the identity $\mathcal{L}_u h^{\mu\nu} = -h^{\mu\alpha} h^{\nu\beta} \mathcal{L}_u \gamma_{\alpha\beta} - 2\mathcal{A}^{(\mu} v^{\nu)}$, showing that the purely temporal part of $\mathcal{L}_u h^{\mu\nu}$ vanishes, and that its mixed temporal-spatial part depends on $\mathcal{A}^\mu$.

where $\theta := \Theta_{\mu\nu} h^{\mu\nu}$.^{12,13}

As we fix the Coriolis field with respect to u^μ with $(\nabla_{[\mu} u^\alpha) \gamma_{\nu]\alpha} = \kappa_{\mu\nu}$, this implies the constraint

$$u^\beta \gamma_{\alpha[\mu} C^\alpha_{\nu]\beta} = 0. \quad (\text{A.9})$$

Therefore, the term $\tau_\mu \tau_\nu \mathcal{A}^\alpha$ is forbidden. Hence, the connection has the general form¹⁴

$$\begin{aligned} \Gamma_{\mu\nu}^\alpha &= \check{\Gamma}_{\mu\nu}^\alpha + 2\tau_{(\mu} \kappa_{\nu)}^\alpha \\ &+ c_1 \gamma_{(\mu}^\alpha \mathcal{A}_{\nu)} + c_2 \gamma_{\mu\nu} \mathcal{A}^\alpha + c_3 u^\alpha \tau_{(\mu} \mathcal{A}_{\nu)} + c_4 \gamma_{[\mu}^\alpha \mathcal{A}_{\nu]} + c_5 u^\alpha \tau_{[\mu} \mathcal{A}_{\nu]} \\ &+ \left(d_1 \gamma_{(\mu}^\alpha \tau_{\nu)} + d_2 \gamma_{\mu\nu} u^\alpha + d_3 u^\alpha \tau_\mu \tau_\nu + d_4 \gamma_{[\mu}^\alpha \tau_{\nu]} \right) \theta \\ &+ d_5 \Theta_{(\mu}^\alpha \tau_{\nu)} + d_6 \Theta_{\mu\nu} u^\alpha + d_7 \Theta_{[\mu}^\alpha \tau_{\nu]}, \end{aligned} \quad (\text{A.10})$$

where the c_i and d_i are constants. The non-metricities and torsion of this connection can be computed to be

$$Q_{\mu\nu} = -(1 - c_5) \mathcal{A}_{[\mu} \tau_{\nu]} - c_3 \mathcal{A}_{(\mu} \tau_{\nu)} - (d_2 \gamma_{\mu\nu} + d_3 \tau_\mu \tau_\nu) \theta - d_6 \Theta_{\mu\nu}, \quad (\text{A.11a})$$

$$\begin{aligned} Q_\alpha^{\mu\nu} &= (c_1 - c_4) \mathcal{A}_\alpha h^{\mu\nu} + (-1 + c_3 + c_5) \tau_\alpha u^{(\mu} \mathcal{A}^{\nu)} + (c_1 + 2c_2 + c_4) \gamma^{(\mu}{}_\alpha \mathcal{A}^{\nu)} \\ &+ \left[(d_1 - d_4) \tau_\alpha h^{\mu\nu} + 2d_2 \gamma^{(\mu}{}_\alpha u^{\nu)} \right] \theta + (d_5 - d_7) \tau_\alpha \Theta^{\mu\nu} + 2d_6 \Theta^{(\mu}{}_\alpha u^{\nu)}, \end{aligned} \quad (\text{A.11b})$$

$$T^\alpha{}_{\mu\nu} = 2c_4 \gamma_{[\mu}^\alpha \mathcal{A}_{\nu]} + 2c_5 u^\alpha \tau_{[\mu} \mathcal{A}_{\nu]} + 2d_4 \delta_{[\mu}^\alpha \tau_{\nu]} \theta + 2d_7 \Theta_{[\mu}^\alpha \tau_{\nu]}, \quad (\text{A.11c})$$

where we have used $\gamma_{[\mu}^\alpha \tau_{\nu]} = \delta_{[\mu}^\alpha \tau_{\nu]}$.

Now we consider how the different quantities appearing in (A.8) transform under Galilean boosts $u^\mu \rightarrow u^\mu - v^\mu$. We denote the *linearised* change of any quantity $f[u]$ under this boost by $\delta_v f[u]$, i.e.,

$$f[u - v] =: f[u] + \delta_v f[u] + (\text{terms at least quadratic in } v).^{15} \quad (\text{A.12})$$

Using this notation, we have

$$\delta_v u^\mu = -v^\mu, \quad \delta_v \gamma_{\mu\nu} = 2\tau_{(\mu} v_{\nu)}, \quad \delta_v \gamma^\mu{}_\nu = v^\mu \tau_\nu, \quad \delta_v \mathcal{A}_\mu = \tau_\mu (\mathcal{A} \cdot v), \quad (\text{A.13a})$$

$$\delta_v \Theta_{\mu\nu} = v_{(\mu} \mathcal{A}_{\nu)} - 2\tau_{(\mu} v_{\nu)} \omega_{\alpha} u^\alpha - \frac{1}{2} \mathcal{L}_v \gamma_{\mu\nu}, \quad \delta_v \theta = \mathcal{A} \cdot v - \frac{1}{2} h^{\mu\nu} \mathcal{L}_v \gamma_{\mu\nu}, \quad (\text{A.13b})$$

¹²A similar approach was followed in ref. [33] to define ‘reduced non-metricities’ and ‘reduced torsion’ in order to remove the appearance of $\omega_{\mu\nu}$ in the identities (3.11). However, terms with $\Theta_{\mu\nu}$ were not considered. Furthermore, a term of the form ‘ $\mathcal{A}_\alpha h^{\mu\nu}$ ’ was omitted when listing the possible terms in the non-metricity $Q_\alpha^{\mu\nu}$ (see Appendix A.1 in ref. [33]). As a result, the proposed construction in that paper is not exhaustive and therefore not compatible with the study of the present paper.

¹³Replacing $\Theta_{\mu\nu}$ by $\sigma_{\mu\nu} := \Theta_{\mu\nu} - \frac{\theta}{3} \gamma_{\mu\nu}$ is also possible, but equivalent. However, using the former is simpler for the proof.

¹⁴The terms $\gamma_{\mu\nu} u^\alpha$ and $\Theta_{\mu\nu} u^\alpha$ are not dimensionless, contrary to the other terms in (A.10). Therefore, the coefficients d_2 and d_6 must have the dimension of a velocity to the power -2 .

¹⁵In fact, for the quantities considered in the following, only quadratic terms would appear.

where we defined

$$v_\mu := \gamma_{\mu\nu} v^\nu, \quad v_{\mu\nu} := \partial_{[\mu} v_{\nu]}, \quad \mathcal{A} \cdot v := \gamma_{\mu\nu} \mathcal{A}^\mu v^\nu. \quad (\text{A.14})$$

For the connection to be boost-invariant, by [Proposition A.1](#) it is necessary that its non-metricities and torsion are boost-invariant; in particular, their linear boost variations $\delta_v Q_{\mu\nu}$, $\delta_v Q_{\alpha}{}^{\mu\nu}$, $\delta_v T^\alpha{}_{\mu\nu}$ need to be zero. Additionally, because these need to vanish for *any* spacelike vector field v^μ , if two terms appearing in one of these expressions differ as functionals of v^μ , then they need to vanish independently. Using the linearised boost, rather than the full boost, will simplify the computations, and, as we will see below, this will be sufficient to prove [Proposition 3.3](#).

First computing the linear boost variation of $Q_{\mu\nu}$ as given by [\(A.11a\)](#), using [\(A.13\)](#) we obtain

$$\begin{aligned} \delta_v Q_{\mu\nu} &= -(1 - c_5) (\delta_v \mathcal{A}_{[\mu} \tau_{\nu]}) - c_3 (\delta_v \mathcal{A}_{(\mu} \tau_{\nu)}) - d_2 (\delta_v \gamma_{\mu\nu}) \theta \\ &\quad - (d_2 \gamma_{\mu\nu} + d_3 \tau_\mu \tau_\nu) \delta_v \theta - d_6 \delta_v \Theta_{\mu\nu} \\ &= -(c_3 + d_3) (\mathcal{A} \cdot v) \tau_\mu \tau_\nu - 2d_2 \tau_{(\mu} v_{\nu)} \theta - d_2 (\mathcal{A} \cdot v) \gamma_{\mu\nu} \\ &\quad + (d_2 \gamma_{\mu\nu} + d_3 \tau_\mu \tau_\nu) \frac{1}{2} h^{\rho\sigma} \mathcal{L}_v \gamma_{\rho\sigma} - d_6 (v_{(\mu} \mathcal{A}_{\nu)} - 2\tau_{(\mu} v_{\nu)} \omega_{\nu)} u^\alpha - \frac{1}{2} \mathcal{L}_v \gamma_{\mu\nu}. \end{aligned} \quad (\text{A.15})$$

This must vanish for all spacelike vector fields v^μ . Considering individually the coefficients of terms depending functionally on v^μ in different ways, this imposes

- $d_2 = 0$ as coefficient of $\tau_{(\mu} v_{\nu)} \theta$,
- $d_3 = 0$ as coefficient of $\tau_\mu \tau_\nu \frac{1}{2} h^{\rho\sigma} \mathcal{L}_v \gamma_{\rho\sigma}$,
- $d_6 = 0$ as coefficient of $\frac{1}{2} \mathcal{L}_v \gamma_{\mu\nu}$, and finally
- $c_3 + d_3 = 0$ as coefficient of $(\mathcal{A} \cdot v) \tau_\mu \tau_\nu$, i.e., (with $d_3 = 0$ from above) $c_3 = 0$.

Next, for the variation of $T^\alpha{}_{\mu\nu}$ as given by [\(A.11c\)](#) we obtain

$$\begin{aligned} \delta_v T^\alpha{}_{\mu\nu} &= 2c_4 (\delta_v \gamma^\alpha{}_{[\mu} \mathcal{A}_{\nu]}) + 2c_4 \gamma^\alpha{}_{[\mu} \delta_v \mathcal{A}_{\nu]} + 2c_5 (\delta_v u^\alpha) \tau_{[\mu} \mathcal{A}_{\nu]} + 2c_5 u^\alpha \tau_{[\mu} \delta_v \mathcal{A}_{\nu]} \\ &\quad + 2d_4 \delta^\alpha{}_{[\mu} \tau_{\nu]} \delta_v \theta + 2d_7 h^{\alpha\beta} (\delta_v \Theta_{\beta[\mu} \tau_{\nu]}) \\ &= 2(c_4 - c_5 - \frac{1}{2} d_7) v^\alpha \tau_{[\mu} \mathcal{A}_{\nu]} + 2(c_4 + d_4) (\mathcal{A} \cdot v) \delta^\alpha{}_{[\mu} \tau_{\nu]} \\ &\quad - d_4 \delta^\alpha{}_{[\mu} \tau_{\nu]} h^{\rho\sigma} \mathcal{L}_v \gamma_{\rho\sigma} + d_7 (\mathcal{A}^\alpha v_{[\mu} - h^{\alpha\beta} \mathcal{L}_v \gamma_{\beta[\mu} \tau_{\nu]}) . \end{aligned} \quad (\text{A.16})$$

Again looking at the coefficients of terms depending differently on v^μ , vanishing of this expression imposes

- $d_4 = 0$ as coefficient of $\delta^\alpha{}_{[\mu} \tau_{\nu]} h^{\rho\sigma} \mathcal{L}_v \gamma_{\rho\sigma}$,
- $d_7 = 0$ as coefficient of $h^{\alpha\beta} \mathcal{L}_v \gamma_{\beta[\mu} \tau_{\nu]}$,
- $c_4 - c_5 - \frac{1}{2} d_7 = 0$ as coefficient of $v^\alpha \tau_{[\mu} \mathcal{A}_{\nu]}$, i.e., (with $d_7 = 0$ from above) $c_4 = c_5$, and
- $c_4 + d_4 = 0$ as coefficient of $(\mathcal{A} \cdot v) \delta^\alpha{}_{[\mu} \tau_{\nu]}$, i.e., (with $d_4 = 0$ from above) $c_4 = 0$, further implying $c_5 = 0$.

Finally, using all the already established vanishing of some of the constants c_i and d_i , for the linear boost variation of $Q_\alpha^{\mu\nu}$ as given by (A.11b) we obtain

$$\begin{aligned}\delta_v Q_\alpha^{\mu\nu} &= c_1 (\delta_v \mathcal{A}_\alpha) h^{\mu\nu} - \tau_\alpha (\delta_v u^{(\mu} \mathcal{A}^{\nu)}) + (c_1 + 2c_2) (\delta_v \gamma^{(\mu}{}_\alpha) \mathcal{A}^{\nu)} \\ &\quad + d_1 \tau_\alpha h^{\mu\nu} \delta_v \theta + d_5 \tau_\alpha h^{\mu\rho} h^{\nu\sigma} \delta_v \Theta_{\rho\sigma} \\ &= c_1 (\mathcal{A} \cdot v) \tau_\alpha h^{\mu\nu} + (1 + c_1 + 2c_2 + d_5) \tau_\alpha v^{(\mu} \mathcal{A}^{\nu)} \\ &\quad + d_1 \tau_\alpha h^{\mu\nu} (\mathcal{A} \cdot v - \tfrac{1}{2} h^{\rho\sigma} \mathcal{L}_v \gamma_{\rho\sigma}) - d_5 \tau_\alpha \tfrac{1}{2} h^{\mu\rho} h^{\nu\sigma} \mathcal{L}_v \gamma_{\rho\sigma} .\end{aligned}\tag{A.17}$$

Vanishing of this expression imposes

- $d_1 = 0$ as coefficient of $\tau_\alpha h^{\mu\nu} \tfrac{1}{2} h^{\rho\sigma} \mathcal{L}_v \gamma_{\rho\sigma}$,
- $d_5 = 0$ as coefficient of $\tau_\alpha \tfrac{1}{2} h^{\mu\rho} h^{\nu\sigma} \mathcal{L}_v \gamma_{\rho\sigma}$,
- $c_1 = 0$ as coefficient of $(\mathcal{A} \cdot v) \tau_\alpha h^{\mu\nu}$, and
- $1 + c_1 + 2c_2 + d_5 = 0$ as coefficient of $\tau_\alpha v^{(\mu} \mathcal{A}^{\nu)}$, i.e., (with $d_5 = 0$, $c_1 = 0$ from above) $c_2 = -\tfrac{1}{2}$.

Thus we have shown that for the connection (A.10) to be boost-invariant, we need $c_2 = -\tfrac{1}{2}$ and all the other constants to be zero. This shows uniqueness of boost-invariant connections depending linearly on first derivatives of τ_μ and $h^{\mu\nu}$: any such connection has coefficients

$$\tilde{\Gamma}_{\mu\nu}^\alpha = \check{\Gamma}_{\mu\nu}^\alpha + 2\tau_{(\mu} \kappa_{\nu)}^\alpha - \tfrac{1}{2} \gamma_{\mu\nu} \mathcal{A}^\alpha ,\tag{A.18}$$

with non-metricities and torsion

$$\tilde{Q}_{\mu\nu} = \tau_{[\mu} \mathcal{A}_{\nu]} = \omega_{\mu\nu} , \quad \tilde{Q}_\alpha^{\mu\nu} = -\delta_\alpha^{(\mu} \mathcal{A}^{\nu)} , \quad \tilde{T}_{\mu\nu}^\alpha = 0 .\tag{A.19}$$

Note that these are boost-invariant not only to linear order (as we used in the proof), but exactly. Inserting the non-metricities and torsion into the general transformation law (A.1) for the Coriolis field of boost-invariant connections, a direct computation shows that this simplifies to

$$\kappa_{\mu\nu} \xrightarrow{u \rightarrow u-v} \kappa_{\mu\nu} - \partial_{[\mu} \left(v_{\nu]} + \tau_{\nu]} \tfrac{v^2}{2} \right) ,\tag{A.20}$$

as claimed in the statement of [Proposition 3.3](#).

Conversely, since the non-metricities and torsion are boost-invariant, it follows by [Proposition A.1](#) that the unique connection we derived is indeed boost-invariant, which shows existence. $\square$

Remark A.2. In [Appendix B.1](#), we discuss connections only invariant under restricted boosts, for which $\mathcal{A} \cdot v = 0$. This leaves more freedom for the connection, as several of the constraints on the constants c_i and d_i are not imposed.

B. Boost-invariant connections depending on extra structures

In this appendix we detail three methods to define boost-invariant connections which all require the introduction of extra structures on top of the Galilean structure and a choice of Coriolis field.

B.1. With restricted boosts

When proving the uniqueness of the boost-invariant connection (3.15) in Appendix A, we used at several points that for a generic boost, we have $\mathcal{A} \cdot v \neq 0$. A possibility to obtain another connection than (3.15) is to demand the connection to be only invariant under *restricted Galilean boosts* with $\mathcal{A} \cdot v = 0$, i.e., for which the boost parameter field v^μ is orthogonal to the time dilation vector field.

In this case, repeating the analysis of Appendix A, we obtain the following:

- Vanishing of $\delta_v Q_{\mu\nu}$ still yields $d_2 = d_3 = d_6 = 0$, but we may have $c_3 \neq 0$;
- vanishing of $\delta_v T^\alpha_{\mu\nu}$ still yields $d_4 = d_7 = 0$ and $c_4 = c_5$, but we may have $c_4 \neq 0$; and
- vanishing of $\delta_v Q_\alpha^{\mu\nu}$, where we now need to keep the c_3 and $c_4 = c_5$ terms, still yields $d_1 = d_5 = 0$, but we may have $c_1 \neq 0$, and the last equation between the coefficients becomes $1 + c_1 + 2c_2 - c_3 = 0$.

Combined, we only need

$$c_3 = 1 + c_1 + 2c_2, \quad c_5 = c_4, \quad (\text{B.1})$$

leading to

$$Q_{\mu\nu} = (1 - c_4) \tau_{[\mu} \mathcal{A}_{\nu]} - (1 + c_1 + 2c_2) \mathcal{A}_{(\mu} \tau_{\nu)}, \quad (\text{B.2a})$$

$$Q_\alpha^{\mu\nu} = (c_1 - c_4) \mathcal{A}_\alpha h^{\mu\nu} + (c_1 + 2c_2 + c_4) \delta_\alpha^{(\mu} \mathcal{A}^{\nu)}, \quad (\text{B.2b})$$

$$T^\alpha_{\mu\nu} = 2 c_4 \delta_{[\mu}^\alpha \mathcal{A}_{\nu]}. \quad (\text{B.2c})$$

Since $\mathcal{A}_\mu$ is invariant under restricted boosts, the non-metricities and torsion are invariant under restricted boosts. The transformation of the Coriolis field may be shown (by direct calculation) to be

$$\kappa_{\mu\nu} \xrightarrow{u \rightarrow u-v} \kappa_{\mu\nu} + \left((1 + 2c_2) \mathcal{A}_{[\mu} - \partial_{[\mu} \right) \left(v_{\nu]} + \tau_{\nu]} \frac{v^2}{2} \right). \quad (\text{B.3})$$

Given a timelike vector field u^μ , the restricted Galilean boosts define a preferred (2-dimensional) family of timelike vector fields whose members are all related to u^μ by a restricted boost. Therefore, while in this approach to define a (restricted) boost-invariant connection there is no explicit extra structure, preferred observers are implicitly introduced.

Other restrictions than $\mathcal{A} \cdot v \neq 0$ could be considered. An example is $D^\mu v^\nu = 0$ which defines a family of observers related by a ‘spatially constant’ boost. But it is not clear if this would produce a more general connection than the connection (3.15).

B.2. With a mass gauge field

Another possibility to obtain a boost-invariant connection from $(\tau_\mu, h^{\mu\nu}, u^\mu, \kappa_{\mu\nu})$ and linear in the first derivatives of $(\tau_\mu, h^{\mu\nu})$ is to introduce a (*mass*) *gauge field* / Bargmann form M_μ such that [63–66, 34, 67], [12, Chapter 5]

$$M_\mu \xrightarrow{u \rightarrow u-v} M_\mu + v_\mu + \tau_\mu \frac{v^2}{2} . \quad (\text{B.4})$$

Then, any connection constructed from the fields

$$\bar{h}_{\mu\nu} := \gamma_{\mu\nu} - 2M_{(\mu}\tau_{\nu)}, \bar{u}^\mu := u^\mu + M_\alpha h^{\mu\alpha}, \bar{\mathcal{A}}_\mu := \mathcal{L}_{\bar{u}}\tau_\mu, \bar{\Theta}_{\mu\nu} := \frac{1}{2}\mathcal{L}_{\bar{u}}\bar{h}_{\mu\nu}, \bar{\theta} := h^{\alpha\beta}\bar{\Theta}_{\alpha\beta},$$

which are boost-invariant fields, will be boost-invariant. In this sense, the connection is constructed from $(\tau_\mu, h^{\mu\nu}, M_\mu, u^\mu, \kappa_{\mu\nu})$ and linear in the first derivatives of $(\tau_\mu, h^{\mu\nu}, M_\mu)$.

As an example of such a connection, refs. [63–66, 34] consider the following torsionful compatible connection (note that the convention used for unit timelike vector fields in refs. [63, 64, 66, 34] is sign-opposite to ours)

$$\bar{\Gamma}_{\mu\nu}^\alpha := h^{\alpha\sigma} \left(\partial_{(\mu} \bar{h}_{\nu)\sigma} - \frac{1}{2} \partial_\sigma \bar{h}_{\mu\nu} \right) + \bar{u}^\alpha \partial_\mu \tau_\nu \quad (\text{B.5})$$

$$= \bar{u}^\alpha \bar{\Gamma}_{\mu\nu}^\alpha + 2\tau_{(\mu} \bar{u}^\alpha \kappa_{\nu)\sigma} h^{\alpha\sigma} + \bar{u}^\alpha \tau_{[\mu} \bar{\mathcal{A}}_{\nu]} , \quad (\text{B.6})$$

where $\bar{u}^\alpha \kappa_{\mu\nu} = \tau_{[\mu} \bar{u}^\alpha \gamma_{\nu]\alpha} (\bar{\mathcal{A}}^\alpha + h^{\alpha\beta} \partial_\beta) (-2M_\sigma \bar{u}^\sigma + M_\sigma M_\gamma h^{\sigma\gamma})$.

To get the most general connection in this approach, it is sufficient to replace, in the connection (A.10), the fields $\gamma_{\mu\nu}$, u^μ and $\mathcal{A}_\mu$ by, respectively, the boost-invariant fields $\bar{u}^\mu \gamma_{\mu\nu} = \bar{h}_{\mu\nu} + \tau_\mu \tau_\nu (-2M_\alpha \bar{u}^\alpha + M^2)$, $\bar{u}^\mu$ and $\bar{\mathcal{A}}_\mu$, and choosing the Coriolis field of $\bar{u}^\mu$ to be boost-invariant.¹⁶ (For simplicity, we do not consider terms involving $\bar{\Theta}_{\mu\nu}$ or $\bar{\theta}$.) This leads to

$$\begin{aligned} \Gamma_{\mu\nu}^\alpha &= \bar{u}^\alpha \bar{\Gamma}_{\mu\nu}^\alpha + 2\tau_{(\mu} \bar{u}^\alpha \kappa_{\nu)}^\alpha \\ &+ c_1 \bar{u}^\alpha \gamma_{(\mu} \bar{\mathcal{A}}_{\nu)} + c_2 \bar{u}^\alpha \gamma_{\mu\nu} \bar{\mathcal{A}}^\alpha + c_3 \bar{u}^\alpha \tau_{(\mu} \bar{\mathcal{A}}_{\nu)} + c_4 \bar{u}^\alpha \gamma_{[\mu} \bar{\mathcal{A}}_{\nu]} + c_5 \bar{u}^\alpha \tau_{[\mu} \bar{\mathcal{A}}_{\nu]} . \end{aligned} \quad (\text{B.7})$$

This connection is directly boost-invariant, and therefore the parameters c_i are totally free.

The clear disadvantage of this approach is that it requires the introduction of a new field, i.e., namely the mass gauge field M_μ . A direct consequence of the presence of this field is that a preferred timelike vector field is defined, corresponding to $\bar{u}^\mu$. Therefore this approach is equivalent to considering the connection (A.10) and saying that the vector field u^μ is a fundamental building block of the theory, and that no boost invariance with respect to that vector field should be required. In other words, this approach follows a stronger version of the approach of Appendix B.1, as the Galilean boosts are restricted to be the trivial boost.

¹⁶One could also replace $\gamma_{\mu\nu}$ by $\bar{h}_{\mu\nu}$.

Remark B.1. We could construct a connection of the form (3.15) with a gauge field by choosing $\bar{u}\kappa_{\mu\nu} = 0$, $c_2 = -\frac{1}{2}$ and $c_1 = c_3 = c_4 = c_5 = 0$ in the connection (B.7). In this case, we have $\bar{u}\kappa_{\mu\nu} = -\partial_{[\mu}M_{\nu]}$ in the connection (3.15).

Remark B.2. Even if the Coriolis field is fully constructed from the gauge field, as for the connection (B.5), and therefore the connection is constructed only from $(\tau_\mu, h^{\mu\nu}, u^\mu, M_\mu)$, there is always a preferred observer that can be constructed from M_μ . On the contrary, if one considers only $(\tau_\mu, h^{\mu\nu}, u^\mu, \kappa_{\mu\nu})$, as we do in the present paper, no preferred observer can be constructed from $\kappa_{\mu\nu}$ in general. This is why $(\tau_\mu, h^{\mu\nu}, u^\mu, \kappa_{\mu\nu})$ carries less structure than $(\tau_\mu, h^{\mu\nu}, u^\mu, M_\mu)$. The use of the Coriolis field is further supported by the classification of affine connections with respect to Galilean structures [36], which implies that the Coriolis field is one of the ‘building blocks’ of an affine connection with respect to a Galilean structure.

B.2.1. Choice of connection with a mass gauge field

While the gauge field has the disadvantage of introducing a preferred timelike vector field, the freedom in the connection (B.7) offers several interesting possibilities for choosing the non-metricities and torsion, which is not possible with the unique connection (3.15). In particular, there are two natural choices of connection that can be considered:

- (i) *A torsionful compatible connection:* this implies $c_2 = -c_1$, $c_3 = 0$, $c_4 = c_1$, $c_5 = 1$, where c_1 is free. In this case the connection is not unique.
- (ii) *A torsion-free conformally compatible connection:* this implies $c_1 = 2$, $c_2 = -1$, $c_3 = 1$, $c_4 = 0$, $c_5 = 0$, and the connection is uniquely given by

$$\Gamma_{\mu\nu}^\alpha = \bar{u}\check{\Gamma}_{\mu\nu}^\alpha + 2\tau_{(\mu}\bar{u}\kappa_{\nu)}^\alpha + 2\bar{u}\gamma_{(\mu}^\alpha\bar{\mathcal{A}}_{\nu)} - \bar{u}\gamma_{\mu\nu}\bar{\mathcal{A}}^\alpha + \bar{u}^\alpha\tau_{(\mu}\bar{\mathcal{A}}_{\nu)} \quad (\text{B.8a})$$

or, equivalently, by

$$Q_{\mu\nu} = -\bar{\mathcal{A}}_\mu\tau_\nu, \quad Q_\alpha{}^{\mu\nu} = 2\bar{\mathcal{A}}_\alpha h^{\mu\nu}, \quad T^\alpha{}_{\mu\nu} = 0. \quad (\text{B.8b})$$

Here the conformal 1-form n_μ in (3.9) corresponds to $-\bar{\mathcal{A}}_\mu$.

B.3. With a preferred time function

As discussed in Section 3.1 and shown in ref. [33], if the Galilean structure results from the Galilean limit of a globally hyperbolic Lorentzian structure, then there exists globally a scalar field Ψ and a time function t such that $\tau = e^{-\Psi}dt$, which the Frobenius condition $\tau \wedge d\tau = 0$ alone only implies locally. However, given τ_μ , the field Ψ and the time t are not uniquely defined as can be seen with the transformation $t \rightarrow g(t)$ leading to $\tau = e^{-\Psi + \ln \dot{g}}dt$ and therefore $\Psi \rightarrow \Psi - \ln \dot{g}$. This is in contrast with the classical situation where τ_μ is exact, and therefore a preferred, or *absolute*, time function $\bar{t}$ is defined such that $\tau = d\bar{t}$.

In the approach of this section to construct a boost-invariant connection, we reintroduce a preferred time function $\bar{t}$, and related *time dilation field* $\bar{\Psi}$, such that

$$\tau = e^{-\bar{\Psi}} d\bar{t} . \quad (\text{B.9})$$

Note that, even with the introduction of this ‘preferred’ time function, the physical time is still measured by τ_μ , and therefore the spatial dependence of $\bar{\Psi}$ induces physical time dilation. We define the field $X_\mu := \partial_\mu \bar{\Psi}$, with $\mathcal{A}^\mu = h^{\mu\alpha} X_\alpha$.

Remark B.3. While this approach introduces a 1-form field X_μ as an extra structure, similarly to the gauge field approach with M_μ , these two 1-forms differ in two ways: (i) only the temporal part of X_μ is independent of $\mathcal{A}^\mu$ while M_μ is fully independent of $\mathcal{A}^\mu$, and (ii) X_μ is boost-invariant, while M_μ transforms as $M_\mu \xrightarrow{u \rightarrow u-v} M_\mu + v_\mu + \tau_\mu \frac{v^2}{2}$.

Therefore, compared to the approach of [Section 3.4](#) and the proof of [Appendix A](#) that lead to the unique connection (3.15), on top of $\mathcal{A}^\mu$, we have the additional field $u^\mu X_\mu$ from which we can construct a connection. Then, a connection constructed from $(\bar{\Psi}, \bar{t}, h^{\mu\nu}, u^\mu, \kappa_{\mu\nu})$ and linear in the first derivatives of $(\bar{\Psi}, h^{\mu\nu})$ has the general form¹⁷

$$\begin{aligned} \Gamma_{\mu\nu}^\alpha &= \check{\Gamma}_{\mu\nu}^\alpha + 2\tau_{(\mu} \kappa_{\nu)}^\alpha \\ &+ c_1 \gamma^\alpha_{(\mu} \mathcal{A}_{\nu)} + c_2 \gamma_{\mu\nu} \mathcal{A}^\alpha + c_3 u^\alpha \tau_{(\mu} \mathcal{A}_{\nu)} + c_4 \gamma^\alpha_{[\mu} \mathcal{A}_{\nu]} + c_5 u^\alpha \tau_{[\mu} \mathcal{A}_{\nu]} \\ &+ [c_6 \gamma^\alpha_{(\mu} \tau_{\nu)} + c_7 u^\alpha \gamma_{\mu\nu} + c_8 u^\alpha \tau_\mu \tau_\nu + c_9 \gamma^\alpha_{[\mu} \tau_{\nu]}] u^\sigma \partial_\sigma \bar{\Psi} , \end{aligned} \quad (\text{B.10})$$

where the c_i are constants. The non-metricities and torsion of this connection are

$$Q_{\mu\nu} = -(1 - c_5) \mathcal{A}_{[\mu} \tau_{\nu]} - c_3 \mathcal{A}_{(\mu} \tau_{\nu)} - (c_7 \gamma_{\mu\nu} + c_8 \tau_\mu \tau_\nu) u^\sigma \partial_\sigma \bar{\Psi} , \quad (\text{B.11a})$$

$$\begin{aligned} Q_\alpha{}^{\mu\nu} &= (c_1 - c_4) \mathcal{A}_\alpha h^{\mu\nu} + (-1 + c_3 + c_5) \tau_\alpha u^{(\mu} \mathcal{A}^{\nu)} + (c_1 + 2c_2 + c_4) \gamma^{(\mu}_\alpha \mathcal{A}^{\nu)} \\ &+ [(c_6 - c_9) \tau_\alpha h^{\mu\nu} + 2c_7 \gamma^{(\mu}_\alpha u^{\nu)}] u^\sigma \partial_\sigma \bar{\Psi} , \end{aligned} \quad (\text{B.11b})$$

$$T^\alpha{}_{\mu\nu} = 2c_4 \gamma^\alpha_{[\mu} \mathcal{A}_{\nu]} + 2c_5 u^\alpha \tau_{[\mu} \mathcal{A}_{\nu]} + 2c_9 \delta^\alpha_{[\mu} \tau_{\nu]} u^\sigma \partial_\sigma \bar{\Psi} . \quad (\text{B.11c})$$

With a calculation similar to that performed in [Appendix A](#), we can show that the connection (B.10) is boost-invariant if and only if

$$c_3 = 1 + c_1 + 2c_2 , \quad c_5 = c_4 , \quad c_6 = c_1 , \quad c_7 = 0 , \quad c_8 = 1 + c_1 + 2c_2 , \quad c_9 = c_4 , \quad (\text{B.12})$$

leading to

$$\begin{aligned} \Gamma_{\mu\nu}^\alpha &= \check{\Gamma}_{\mu\nu}^\alpha + 2\tau_{(\mu} \kappa_{\nu)}^\alpha \\ &+ c_1 \gamma^\alpha_{(\mu} X_{\nu)} + c_2 \gamma_{\mu\nu} X^\alpha + (1 + c_1 + 2c_2) u^\alpha \tau_{(\mu} X_{\nu)} + c_4 \delta^\alpha_{[\mu} X_{\nu]} , \end{aligned} \quad (\text{B.13})$$

¹⁷We omit terms involving $\mathcal{L}_u \gamma_{\mu\nu}$ as they must vanish for a boost-invariant connection. Indeed, the presence of $\bar{\Psi}$ does not change the derivation made in [Appendix A](#) for these terms, while it does for the terms involving $\mathcal{A}^\mu$.

and

$$Q_{\mu\nu} = (1 - c_4) \tau_{[\mu} X_{\nu]} - (1 + c_1 + 2c_2) X_{(\mu} \tau_{\nu)} , \quad (\text{B.14a})$$

$$Q_{\alpha}{}^{\mu\nu} = (c_1 - c_4) X_{\alpha} h^{\mu\nu} + (c_1 + 2c_2 + c_4) \delta_{\alpha}^{(\mu} X^{\nu)} , \quad (\text{B.14b})$$

$$T^{\alpha}{}_{\mu\nu} = 2c_4 \delta_{[\mu}^{\alpha} X_{\nu]} . \quad (\text{B.14c})$$

The non-metricities and torsion are boost-invariant since X_{μ} is boost-invariant.

While we introduce a preferred time function from which the clock form is defined, the boost-invariant connection we construct in the present approach does not introduce a preferred timelike vector field, unlike the restricted boost approach from [Appendix B.1](#) or the mass gauge field approach from [Appendix B.2](#). Additionally, since a preferred time is also defined in the classical case, for which $\tau = dt$, but still no preferred observer, this makes the ‘preferred-time-function approach’ the most natural approach to construct a boost-invariant connection after the one considered in [Section 3.4](#).

Remark B.4. To get a connection that is independent of the time $\bar{t}$, i.e., independent of the transformation $\bar{\Psi} \rightarrow \bar{\Psi} + f(t)$, the only possibility is $c_6 = c_7 = c_8 = c_9$, leading again to the unique boost-invariant connection [\(3.15\)](#).

B.3.1. Choice of connection with a preferred time function

As for the mass gauge field approach, the freedom in the connection offers several possibilities for choosing the non-metricities and torsion:

- (i) *A torsionful compatible connection:* this implies $c_1 = 1$, $c_2 = -1$ and $c_4 = 1$, and the connection is uniquely given by

$$\Gamma_{\mu\nu}^{\alpha} = \check{\Gamma}_{\mu\nu}^{\alpha} + 2\tau_{(\mu} \kappa_{\nu)}^{\alpha} + \gamma^{\alpha}{}_{(\mu} X_{\nu)} - \gamma_{\mu\nu} X^{\alpha} + \gamma^{\alpha}{}_{[\mu} X_{\nu]} + u^{\alpha} \tau_{[\mu} X_{\nu]} \quad (\text{B.15a})$$

or, equivalently, by

$$Q_{\mu\nu} = 0 , \quad Q_{\alpha}{}^{\mu\nu} = 0 , \quad T^{\alpha}{}_{\mu\nu} = 2\delta_{[\mu}^{\alpha} X_{\nu]} . \quad (\text{B.15b})$$

- (ii) *A torsion-free conformally compatible connection:* this implies $c_1 = 2$, $c_2 = -1$ and $c_4 = 0$, and the connection is uniquely given by

$$\Gamma_{\mu\nu}^{\alpha} = {}^u\check{\Gamma}_{\mu\nu}^{\alpha} + 2\tau_{(\mu} {}^u\kappa_{\nu)}^{\alpha} + 2{}^u\gamma^{\alpha}{}_{(\mu} X_{\nu)} - {}^u\gamma_{\mu\nu} X^{\alpha} + u^{\alpha} \tau_{(\mu} X_{\nu)} \quad (\text{B.16a})$$

or, equivalently, by

$$Q_{\mu\nu} = -X_{\mu} \tau_{\nu} , \quad Q_{\alpha}{}^{\mu\nu} = 2X_{\alpha} h^{\mu\nu} , \quad T^{\alpha}{}_{\mu\nu} = 0 . \quad (\text{B.16b})$$

As for the mass gauge field approach, the conformal 1-form n_{μ} in [\(3.9\)](#) corresponds to $-\bar{\mathcal{A}}_{\mu}$.

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